Large Data Visualization

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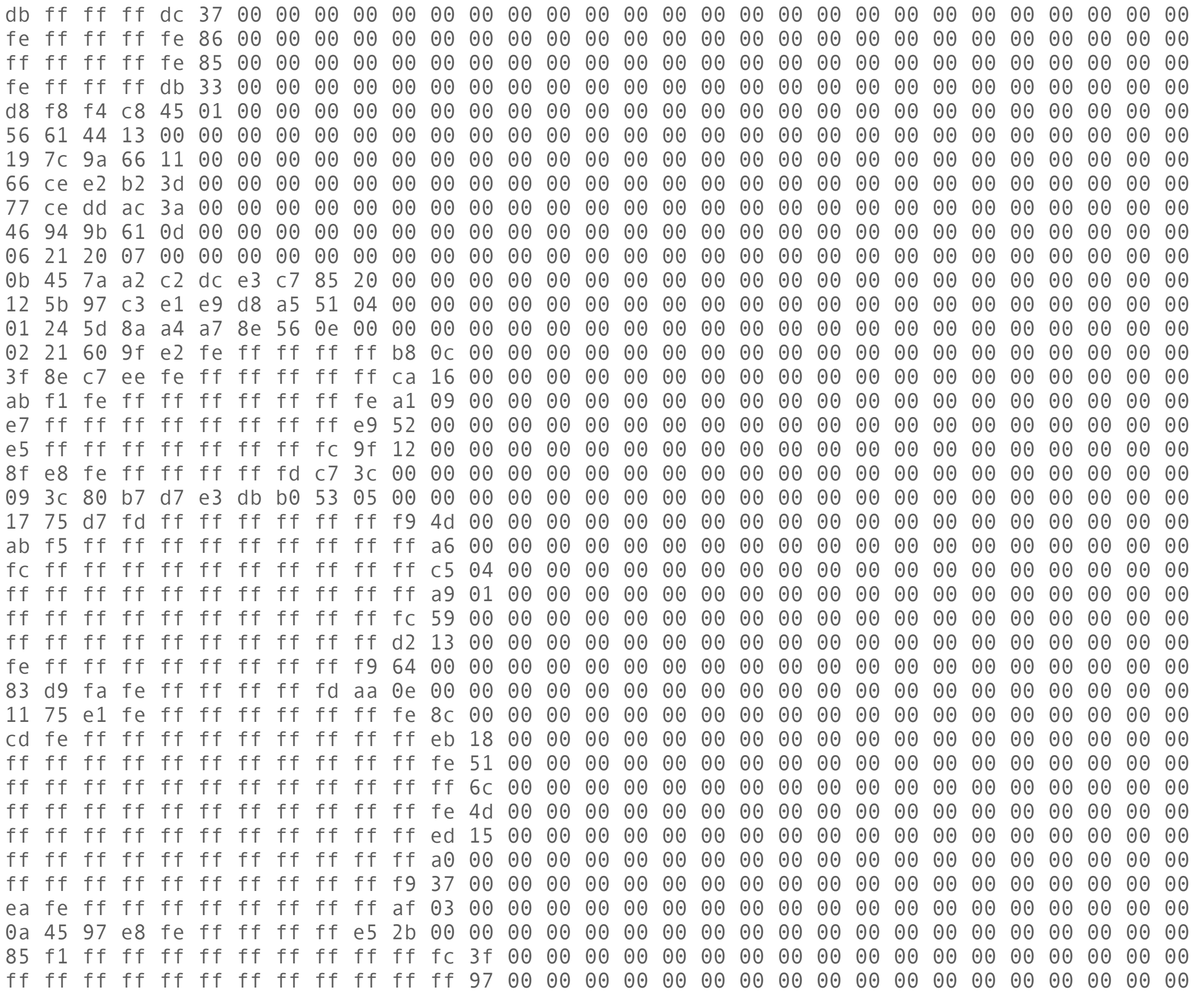
Intro to CS – September 10, 2026

rijksuniversiteit
groningen

What is visualization?

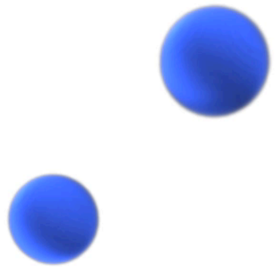
… and why do we do it?

Simulation result (several days on a supercomputer)



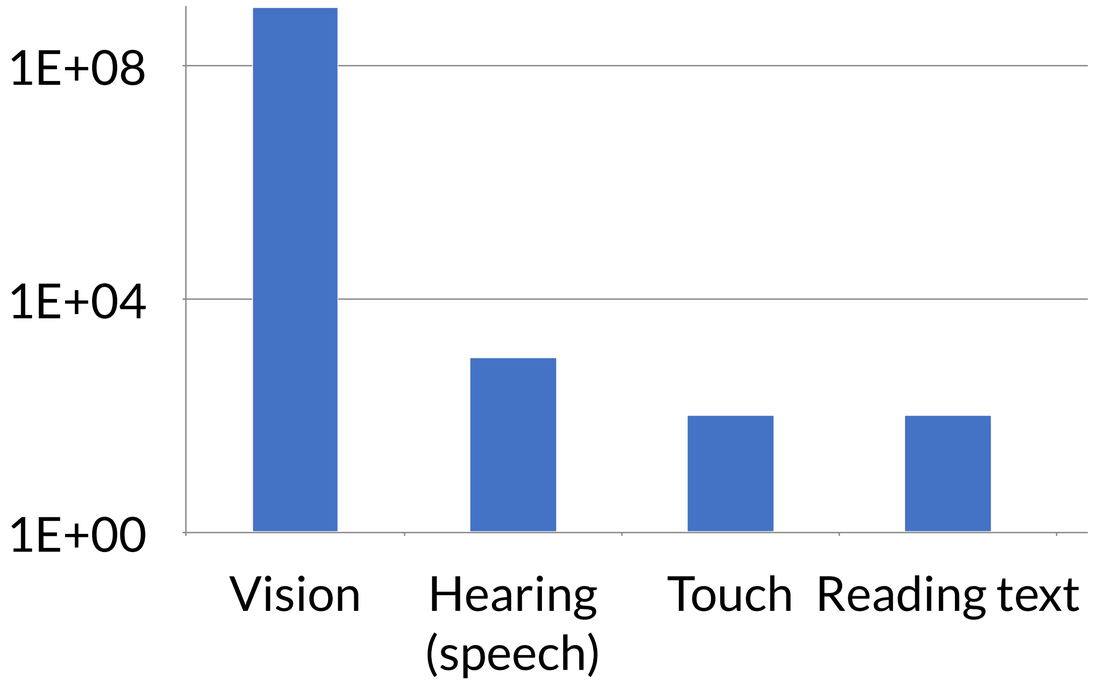
Visualization: Rapid Interpretation of Large Data

The purpose of computing is insight, not numbers [R. Hamming, 1962]



Visual Channel: Our Highest Bandwidth Sense

- **Reading text: ~10^2 bits/s**
  — Reading a book
- **Touch: ~10^2 bits/s**
  — Reading Braille
- **Hearing (speech): ~10^3 bits/s**
  — Listening to conversation
- **Vision: 10^8–10^9 bits/s**
  — Seeing a scene instantly

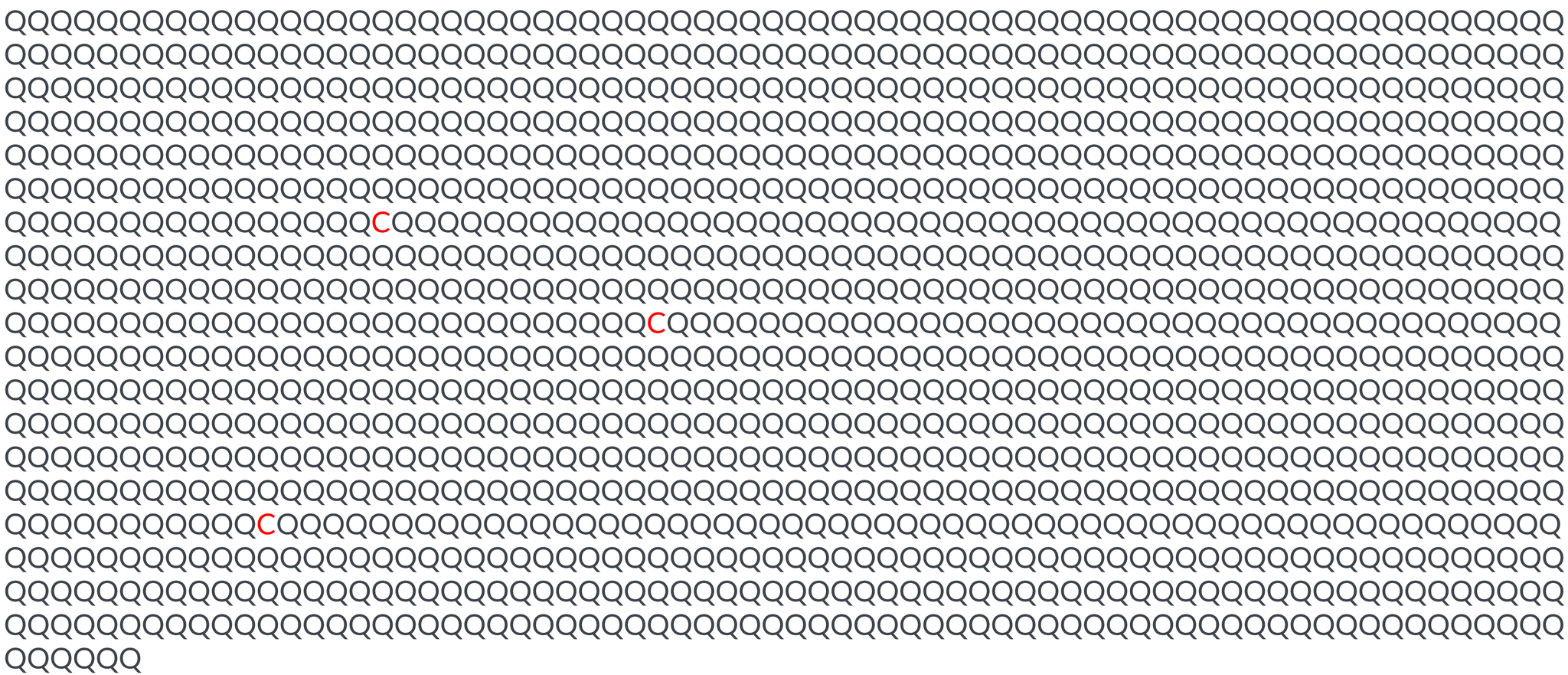


Colin Ware (2020), Fred Attneave (1954), George A. Miller (1956)

Preattentive Processing

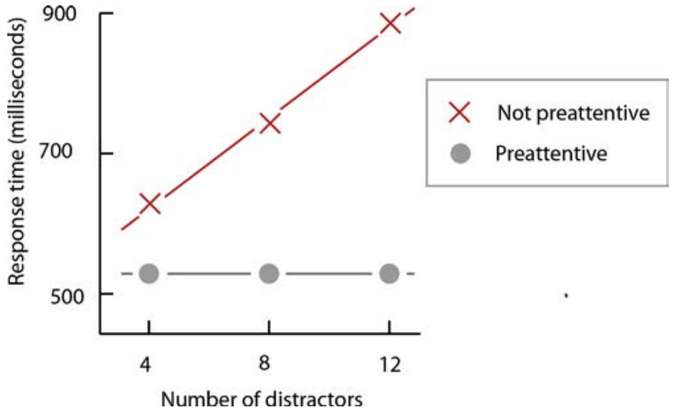
QQQQ

Preattentive Processing

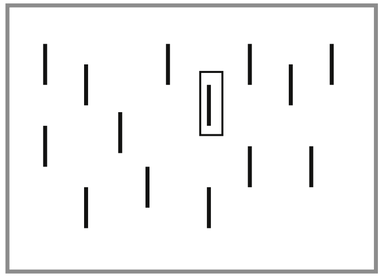

QQQQ

Preattentive Processing

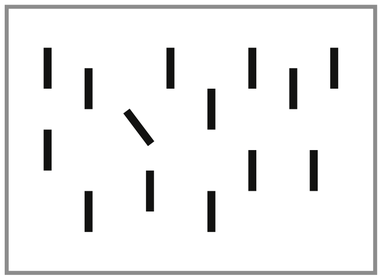
- „Pop-out effect“
- visual properties detected rapidly and in parallel by low-level visual processes



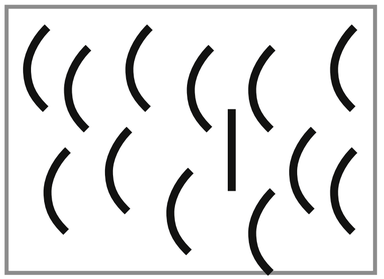
Addition



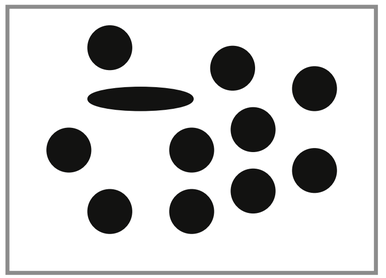
Orientation



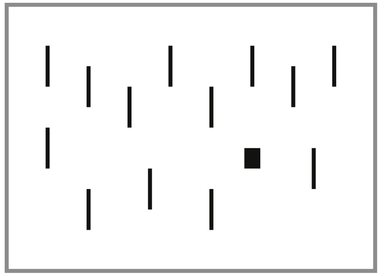
Curved straight



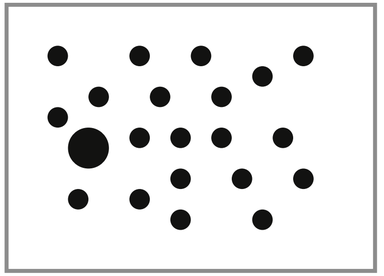
Shape



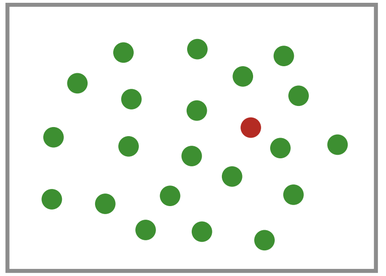
Shape



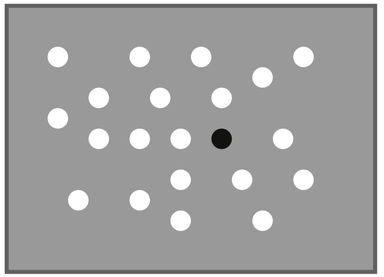
Size



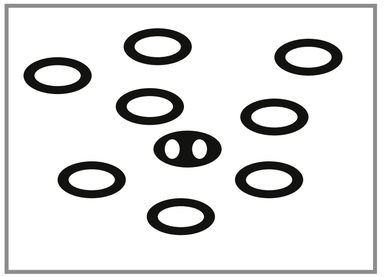
Color



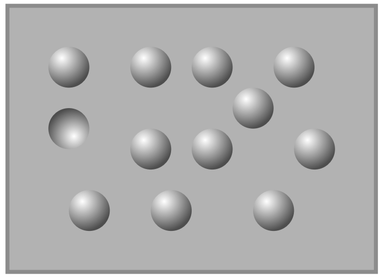
Light/dark



Topology (or count)

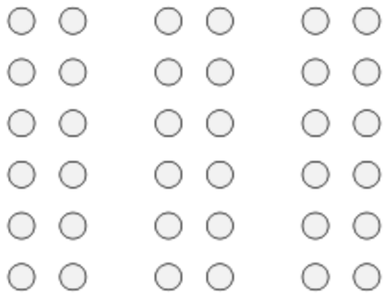


Convex/concave

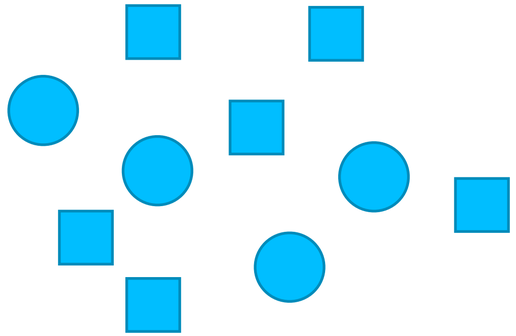


Gestalt Laws for Pattern Perception

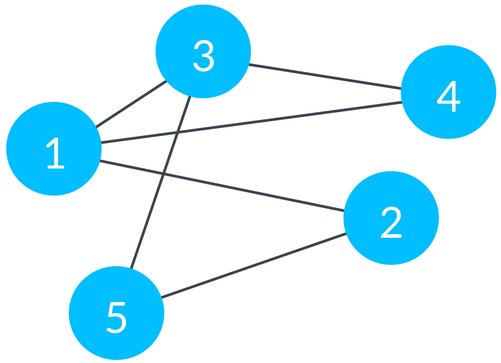
- Human brain excellent at pattern-finding and pattern-association
- Gestalt laws describe when groups of objects are identified



proximity



similarity



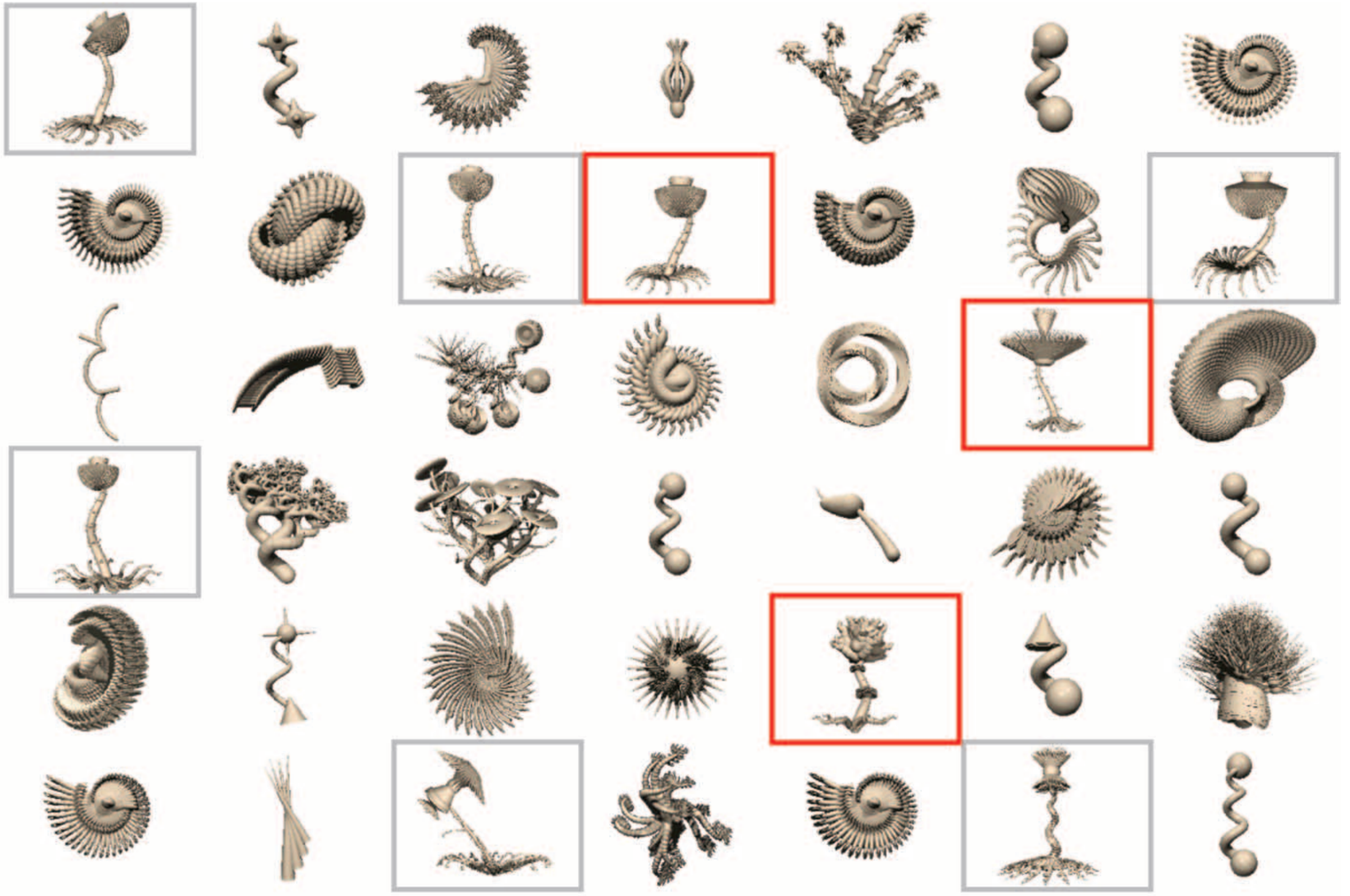
connectedness

Examples of Gestalt laws

Humans learn / generalize from few examples

Example of learning
Alien Language Word “Tufa”:

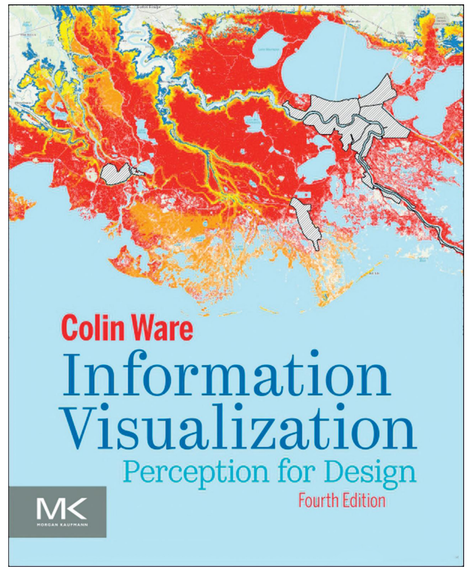
Humans select objects (gray)
based on examples (red)



Joshua B. Tenenbaum et al. ,How to Grow a Mind: Statistics, Structure, and Abstraction.Science331,1279-1285(2011). DOI:10.1126/science.1192788

Visualization Objective

"Design **graphic representations** of data
by taking into account **human sensory capabilities**
in such a way that **important data** elements and data patterns
can be **quickly perceived**"



Colin Ware. Information Visualization (4th ed.). (2021). Elsevier.
https://doi.org/10.1016/c2016-0-02395-1

What is “large” data?

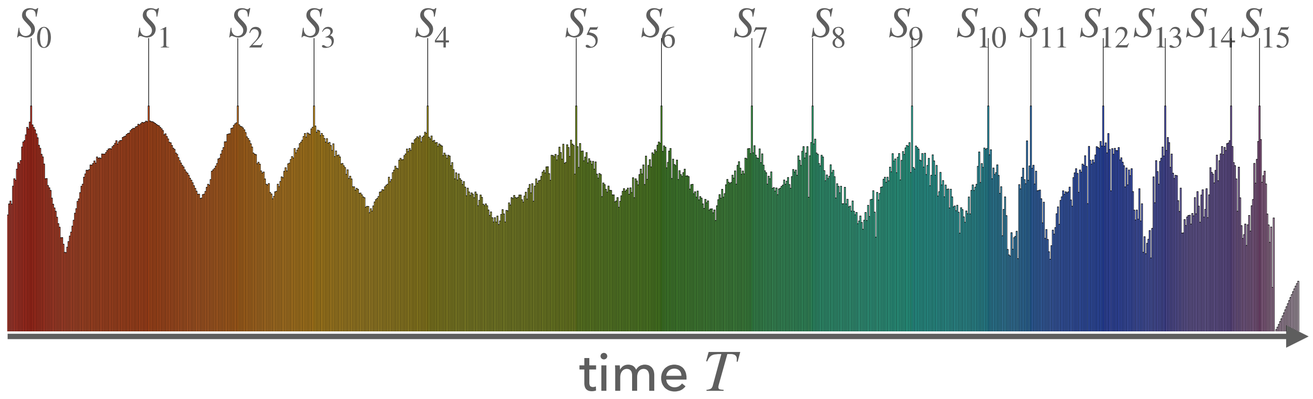
How can we still visualize it?

case 1/3: **high (temporal) resolution**

overlay of selected time steps,
color depicts time

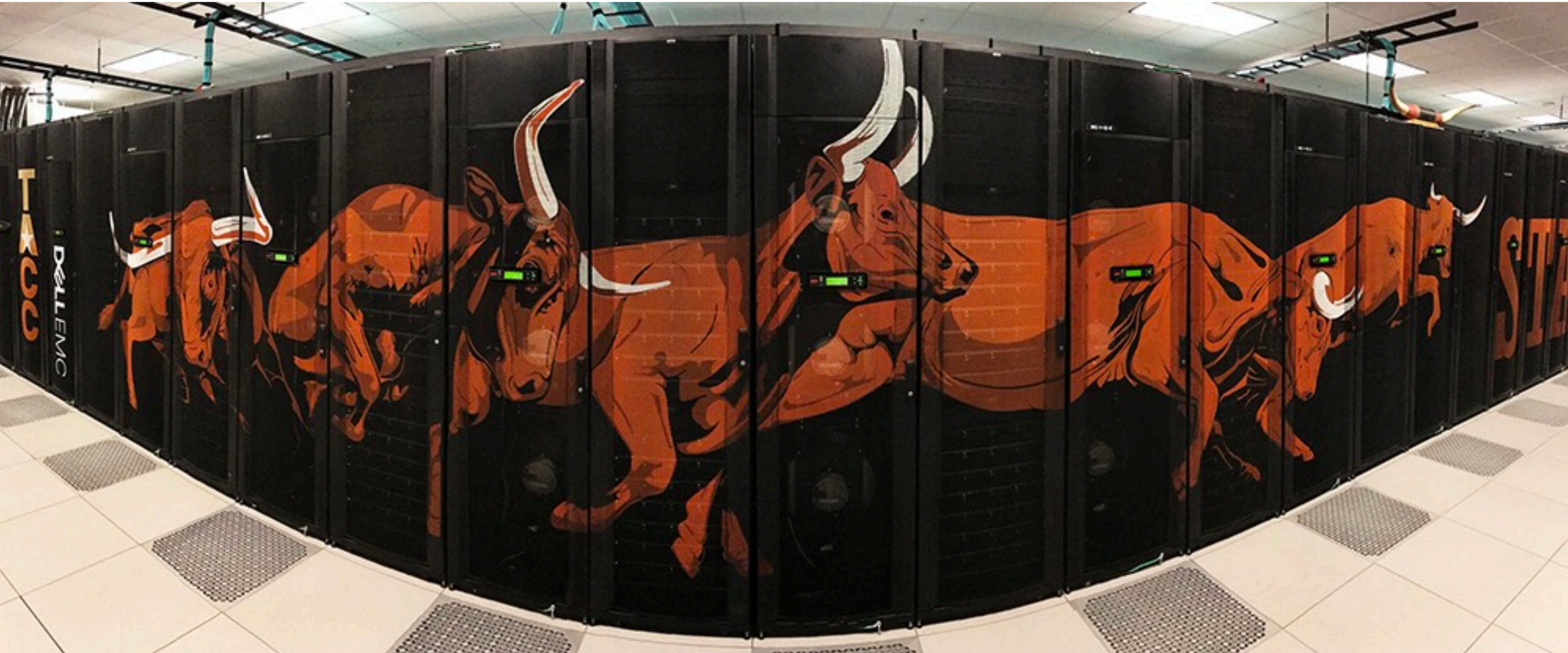
select time steps S

$$S = \arg\min_{S \subset T} \left(\sum_{t \in T} \min \left(d(\lfloor t \rfloor_S, t), d(t, \lceil t \rceil_S) \right) \right)$$

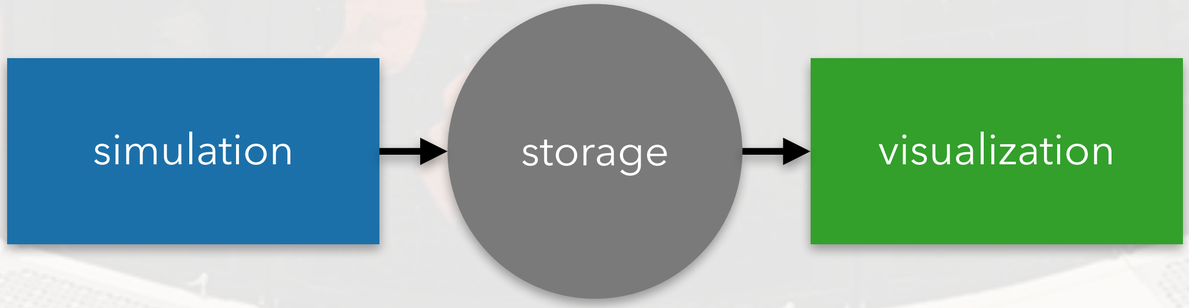


Frey, S., and T. Ertl. "Flow-Based Temporal Selection for Interactive Volume Visualization." *Computer Graphics Forum* 36, no. 8 (2017): 153–65.

case 2/3: **exceeding memory**

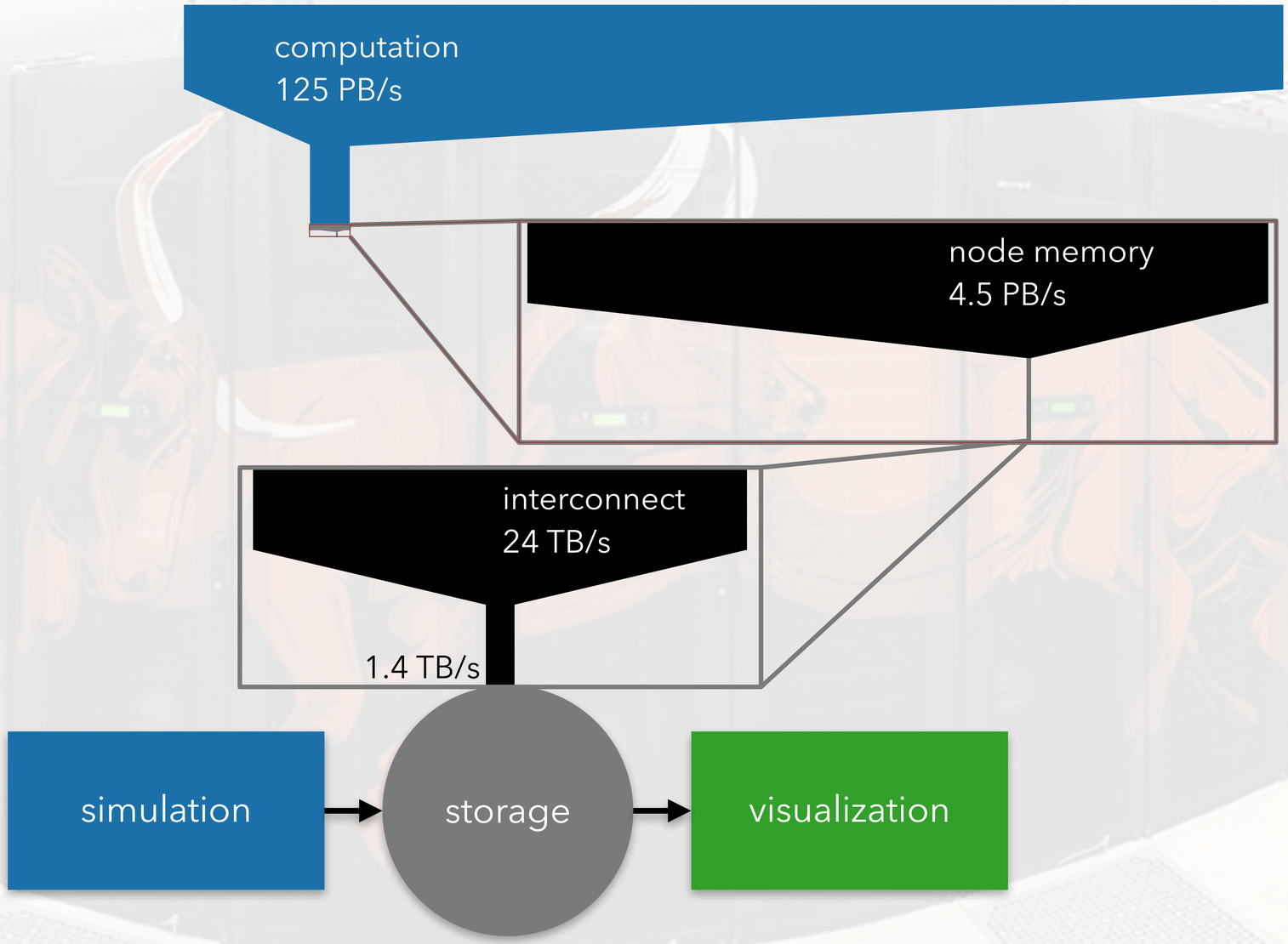


case 2/3: **in situ visualization**



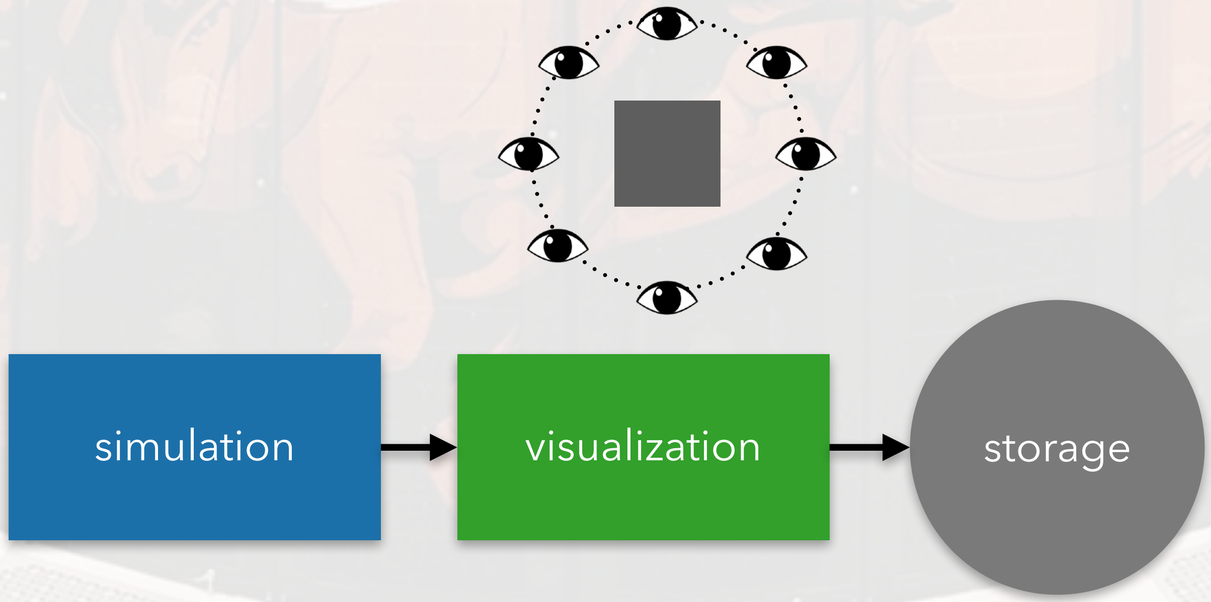
[figure adapted from
Ken Moreland,
Sandia National Laboratories]

case 2/3: **in situ visualization**



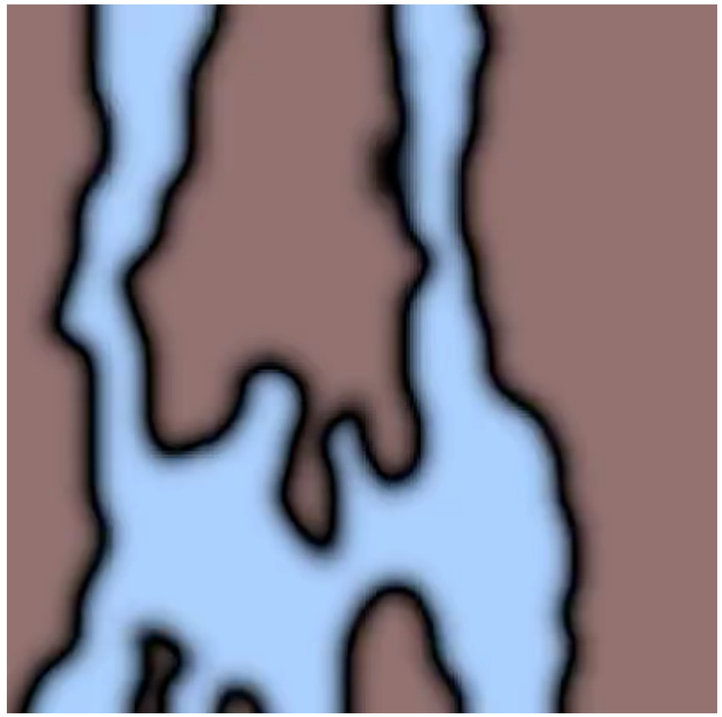
[figure adapted from
Ken Moreland,
Sandia National Laboratories]

case 2/3: **in situ visualization**



[figure adapted from
Ken Moreland,
Sandia National Laboratories]

case 3/3: **many data points**

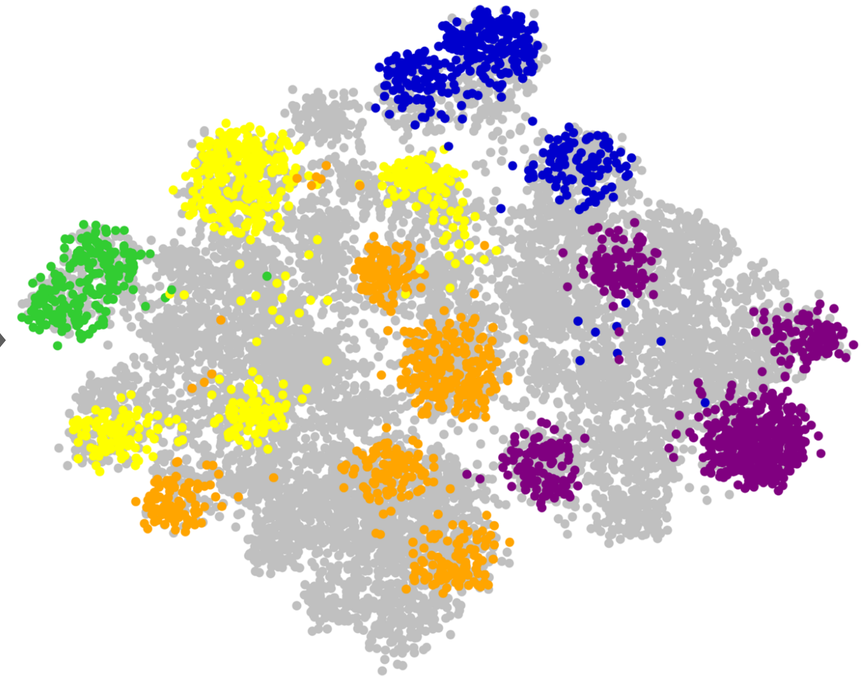


- soil
- channel

S. Reuschen, T. Xu, and W. Nowak, "Bayesian inversion of hierarchical geostatistical models using a parallel-tempering sequential Gibbs MCMC," Advances in Water Resources, 2020.

Dimensionality Reduction (DR)

Preserve similarity relationships of high-dimensional space in low-dimensional space



👍 conveys data distribution

😕 no image content

How to compare many complex data points?

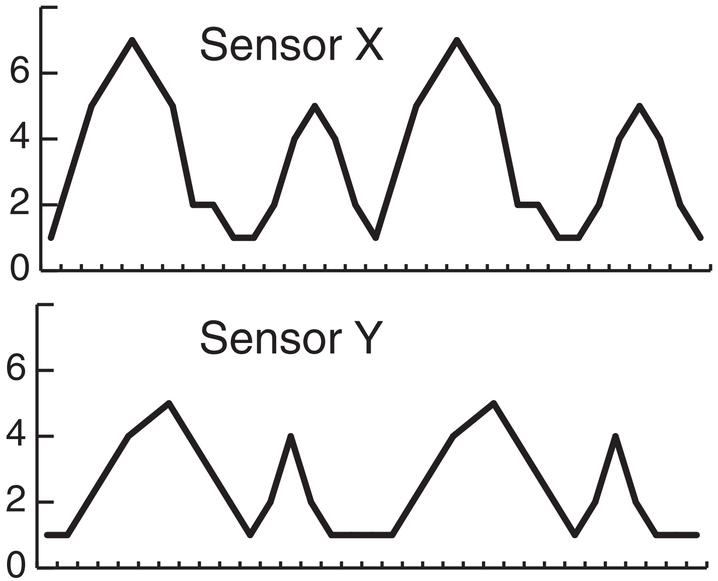
Visually identify clusters, outliers, and patterns in the data

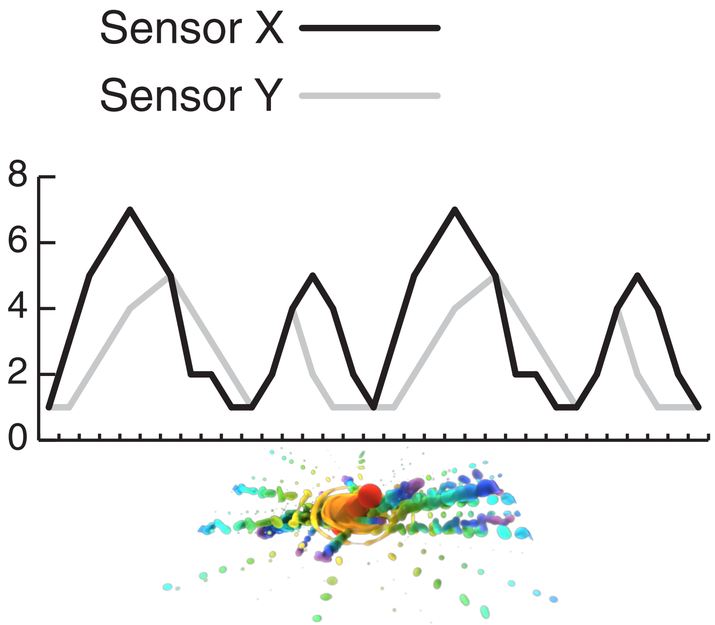
Comparative Visualization: Design Choices

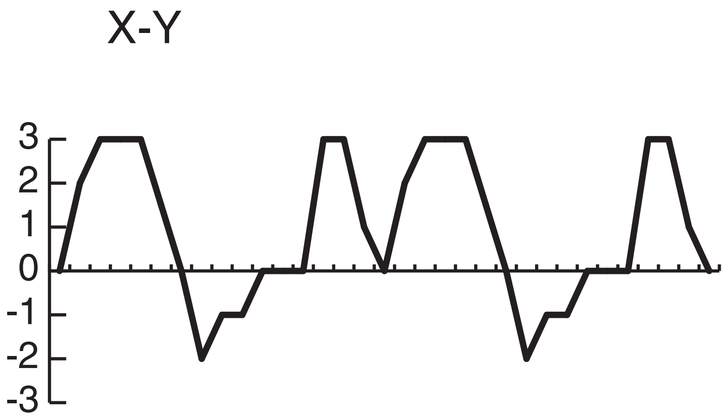
Juxtaposition:
present separately

Superposition:
overlay in same space

Explicit encodings:
directly present relationship







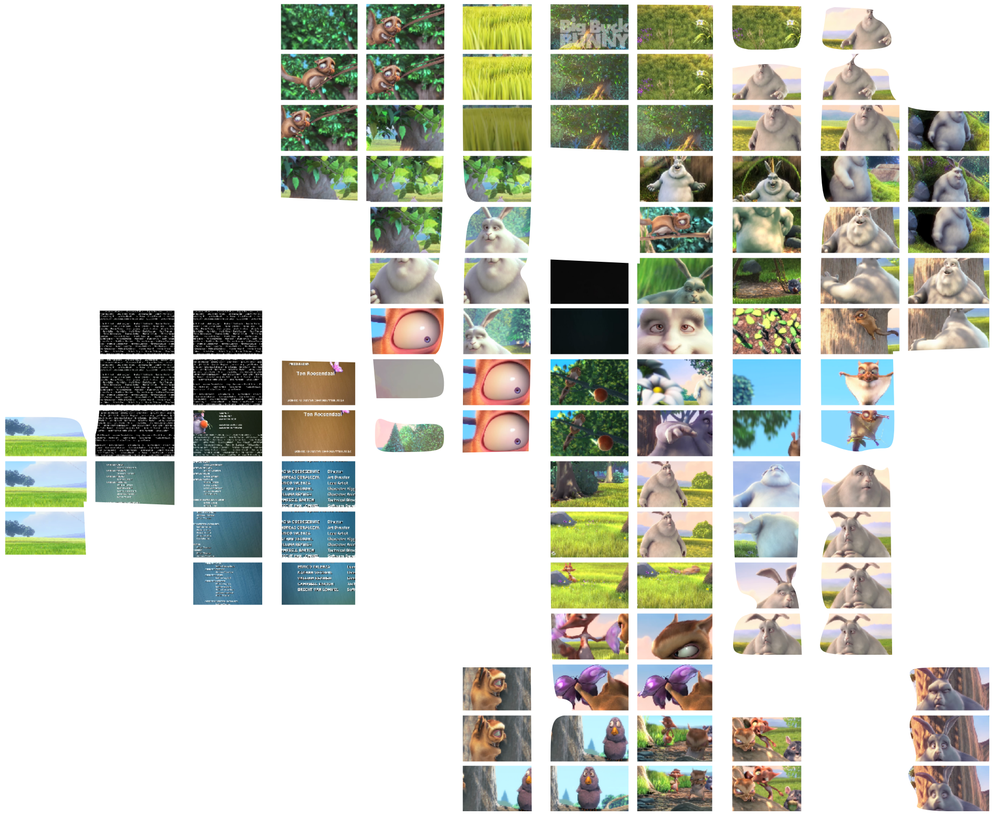
M. Gleicher, D. Albers, R. Walker, I. Jusufi, C. D. Hansen, and J. C. Roberts, "Visual comparison for information visualization," Information Visualization, vol. 10, no. 4, pp. 289–309, Oct. 2011, doi: 10.1177/1473871611416549.

Juxtaposition: image grids

T. Roosendaal. **Big buck bunny.**
In ACM SIGGRAPH ASIA 2008 Computer Animation Festival, 2008.

Organize into Grid Layout

support visual clustering, anomaly detection, exploration



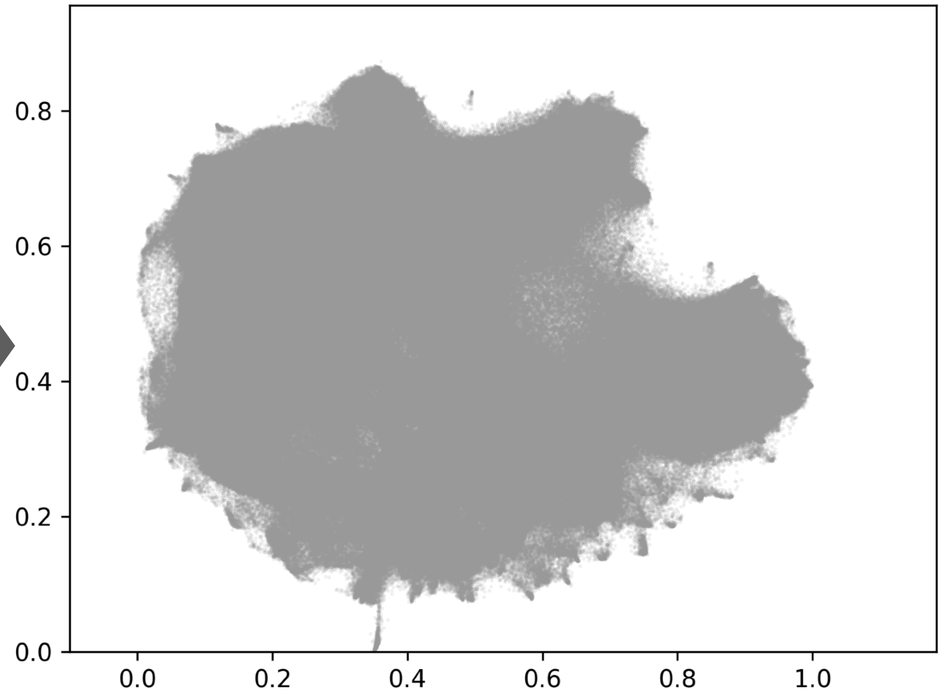
akin to dimensionality reduction:
place similar data points next to each other

Image Grids from DR

ImageNet
(~1.2M points)

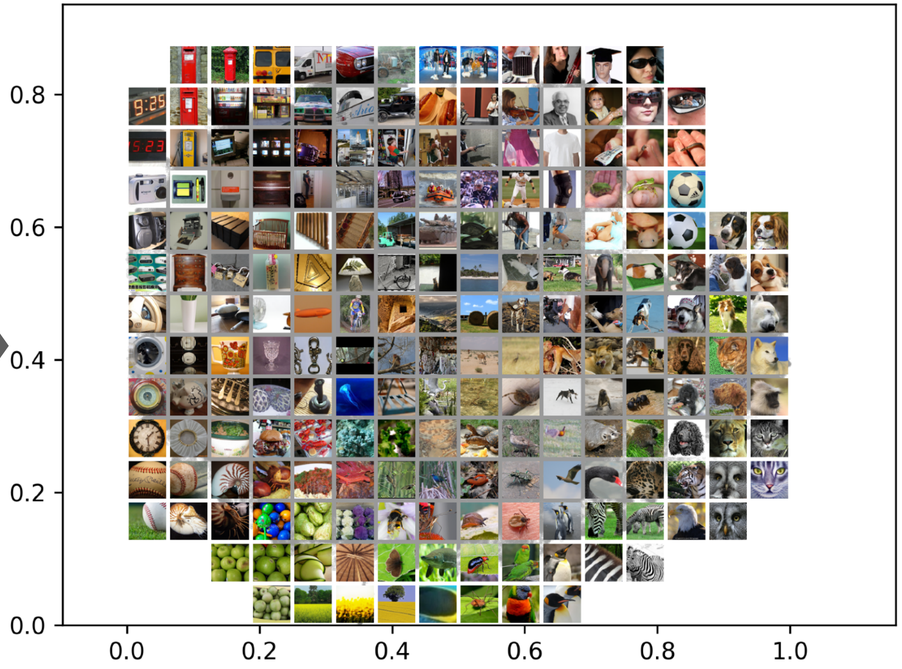
DR

data point
(representing an image)



Dimensionality **R**eduction

LAP



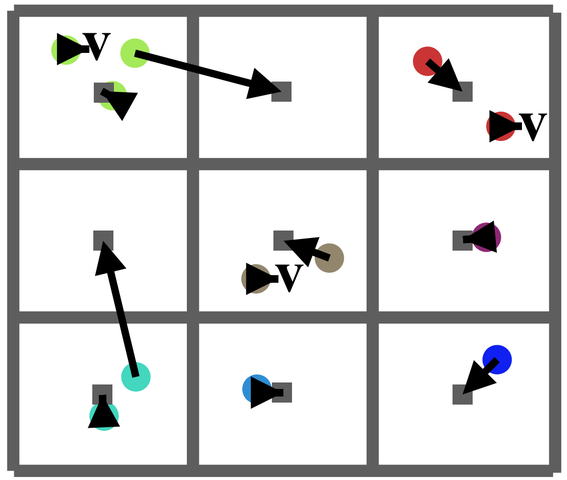
Linear **A**ssignment **P**roblem

S. Frey, ***Sca²Gri: Scalable Gridified Scatterplots***. Computer Graphics Forum, 2025.

Linear Assignment Problem (LAP): points → cells

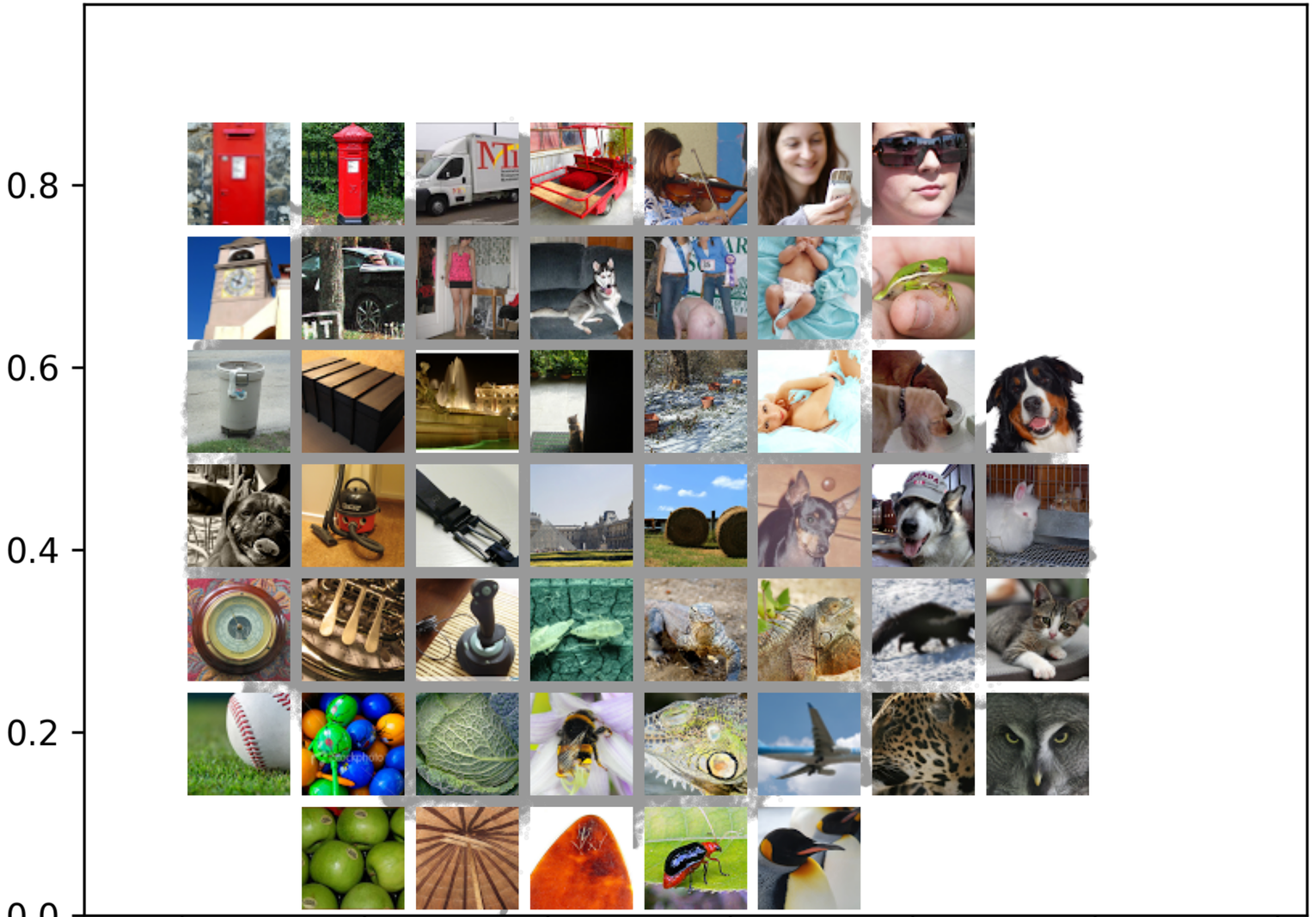
Objective:
minimize sum of all point · → cell □ assignments

Constraint:
| point→cell **|** ≤ displacement threshold



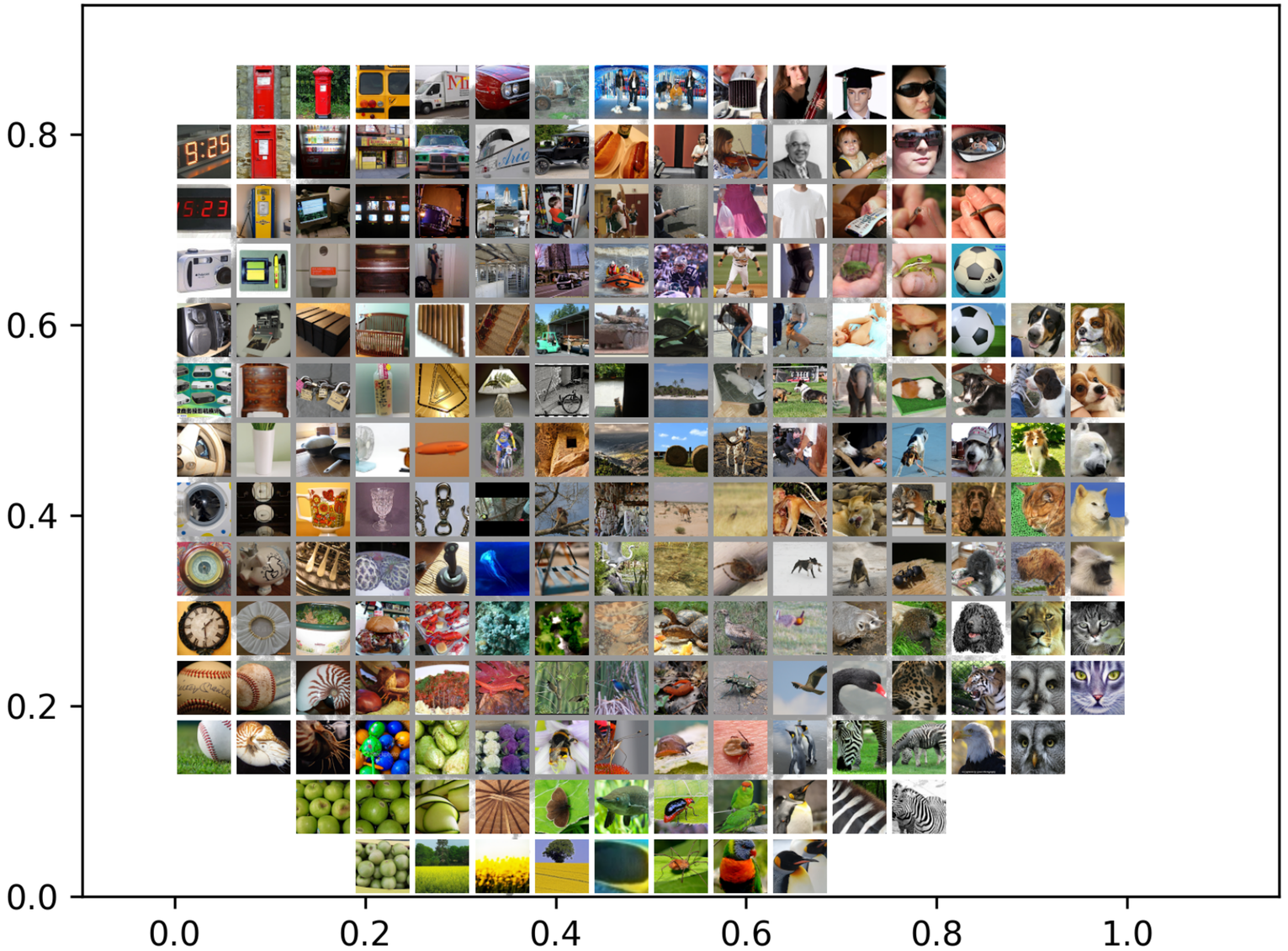
S. Frey, ***Sca²Gri: Scalable Gridified Scatterplots***. Computer Graphics Forum, 2025.

Motivation – ~60 images

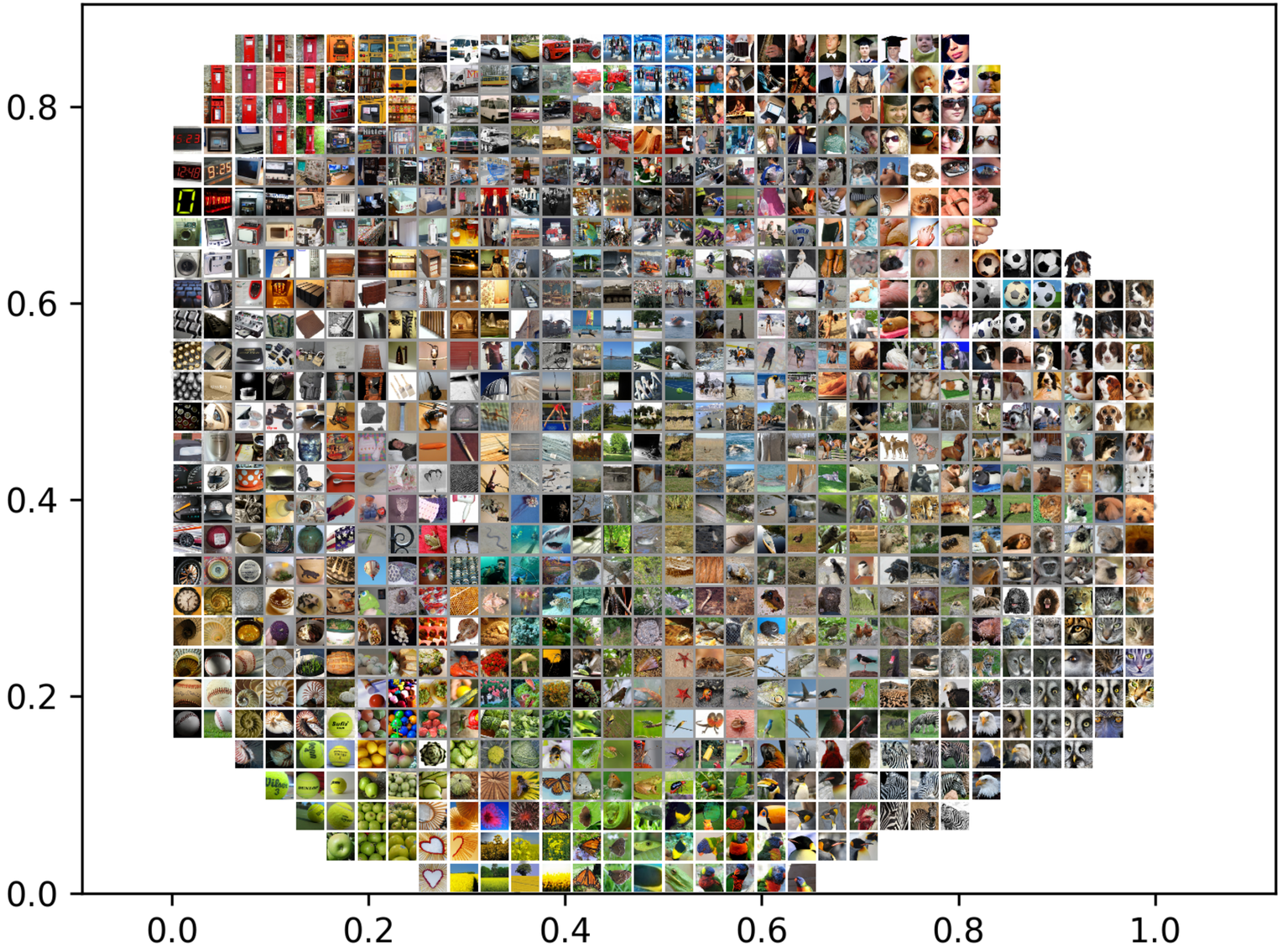


generated by Sca²Gri

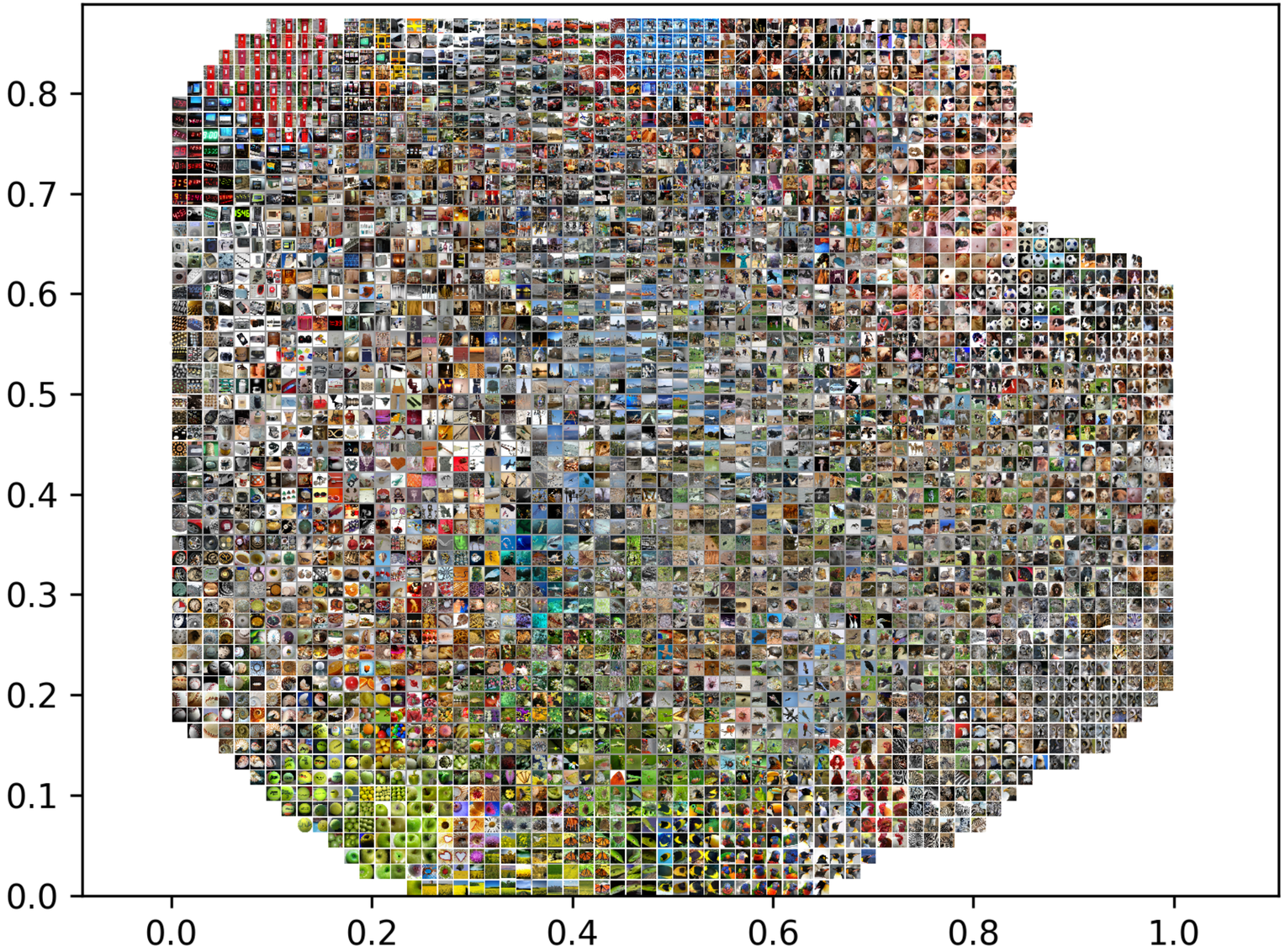
Motivation – ~250 images



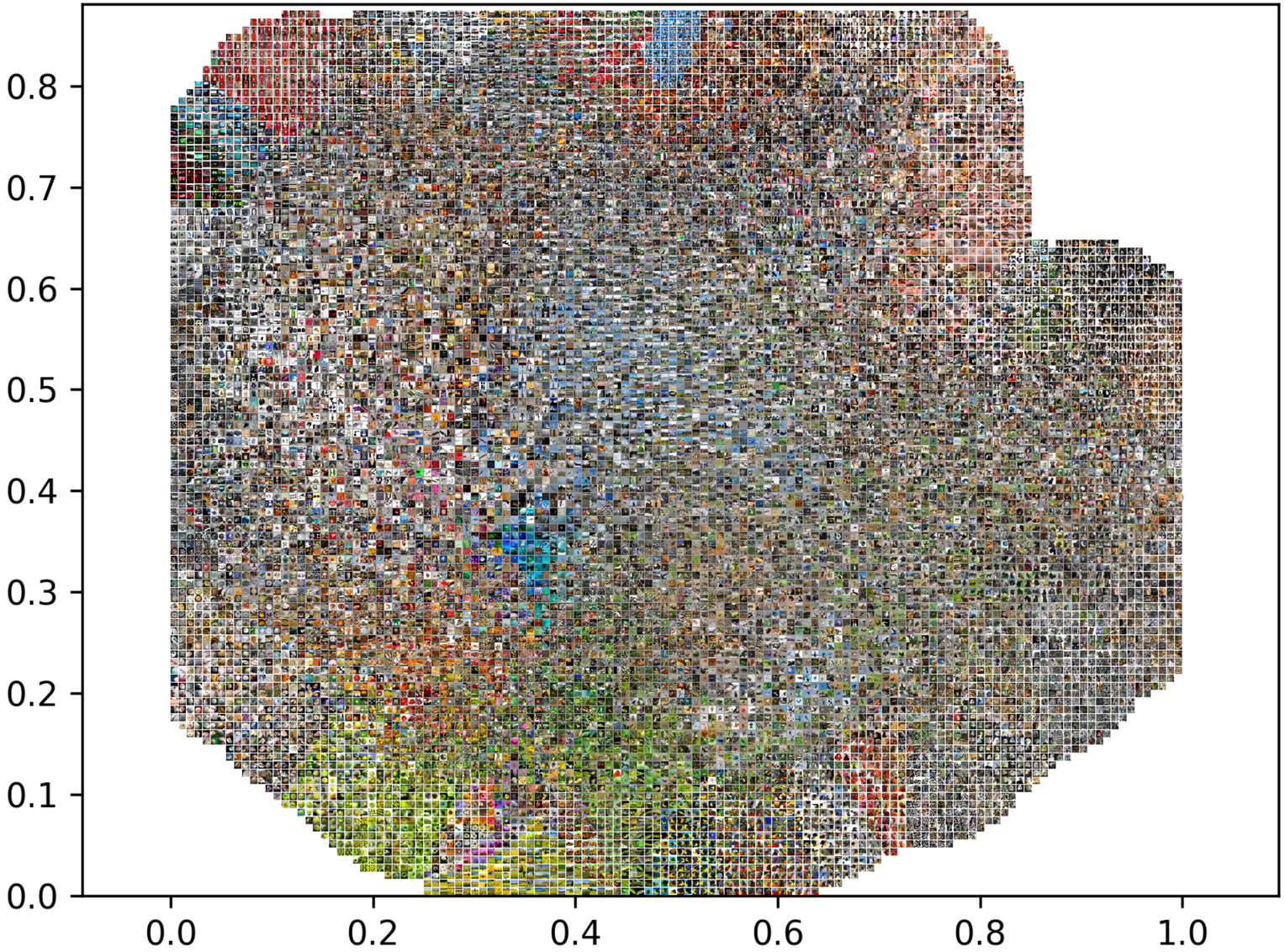
Motivation – ~1000 images



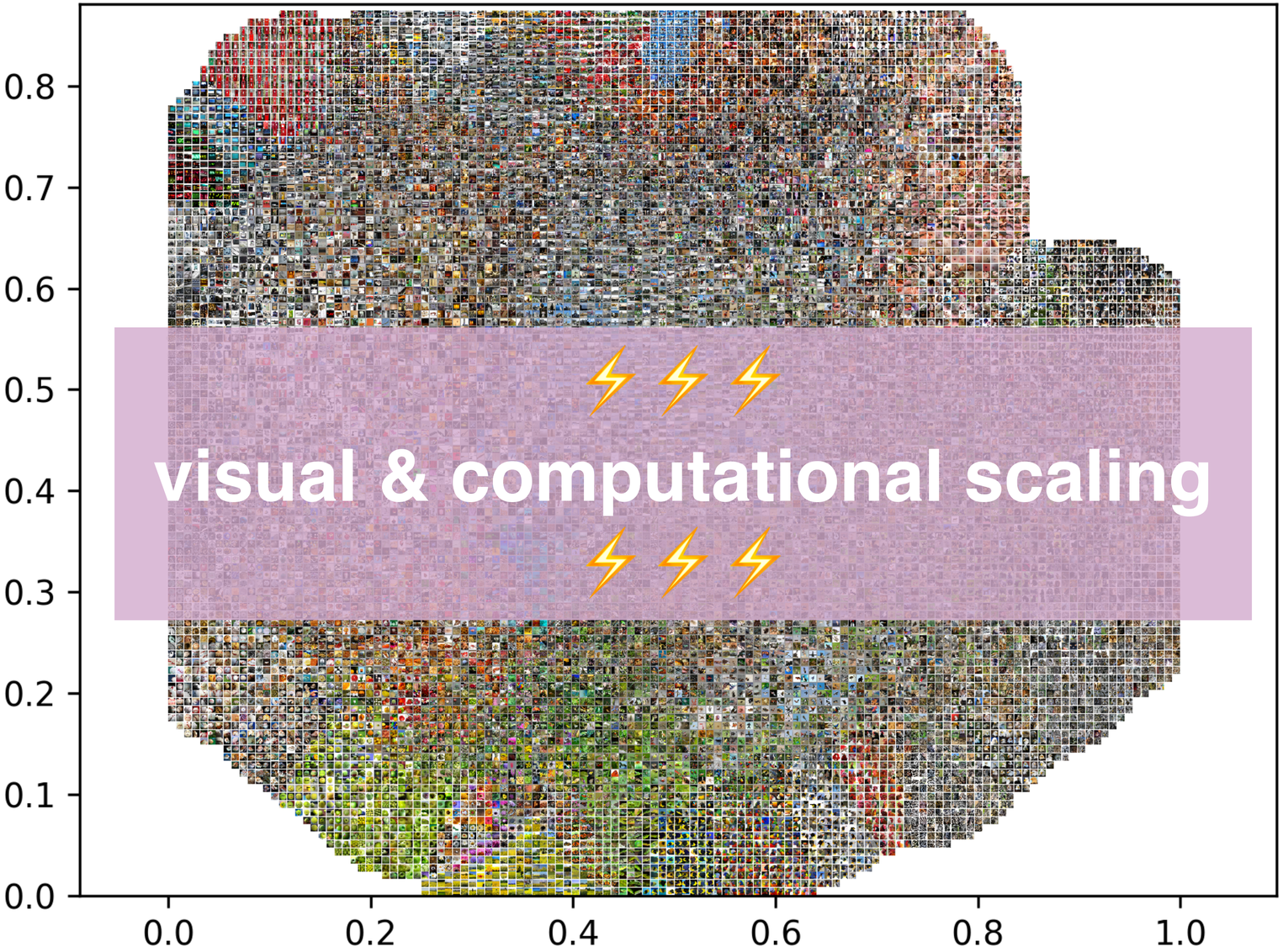
Motivation – ~4000 images



Motivation – ~15000 images

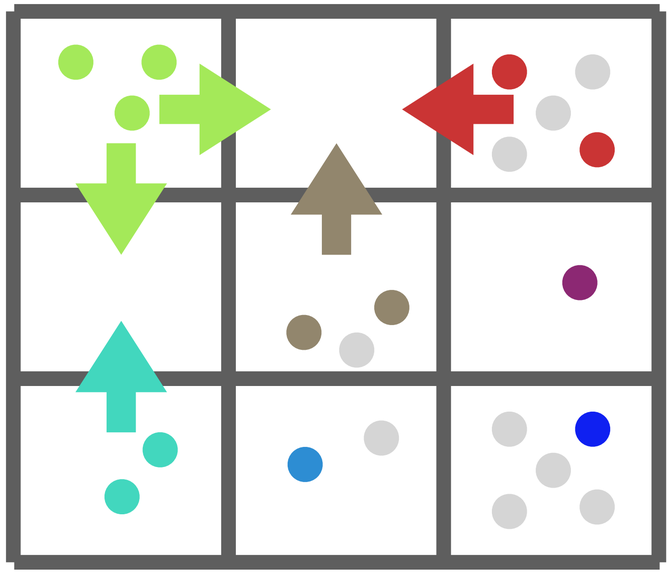


Motivation – ~15000 images



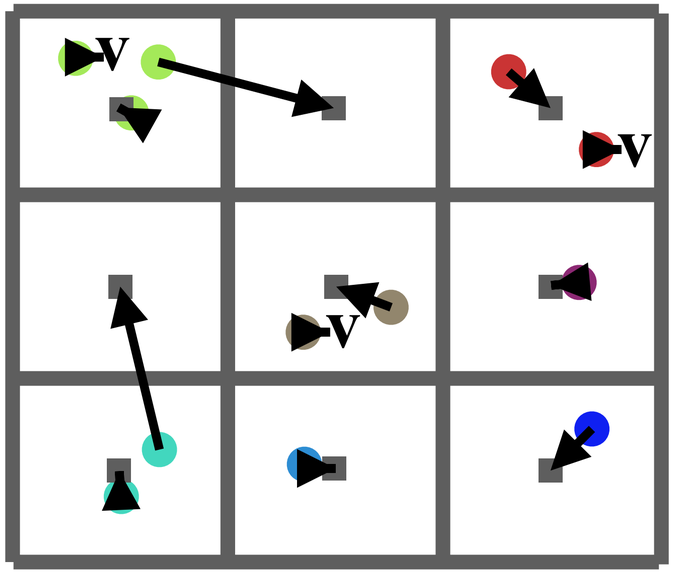
Two-Stage Approach

(I) **Reduce** *#points*



idea: for each cell, only keep as many points as could possibly assigned to other cells (→ constraint & #points >> #cells)

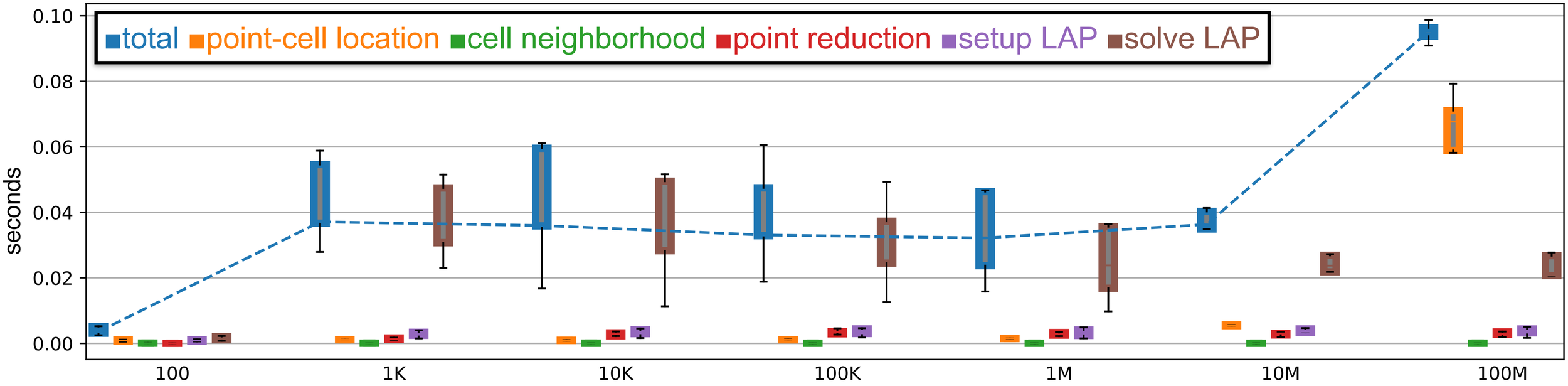
(II) **Assign reduced** points→cells

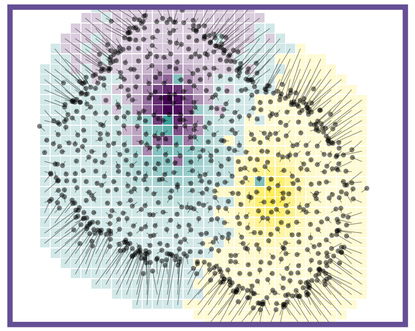


Error for each point→cell assignment **< cell size**

S. Frey, ***Sca²Gri: Scalable Gridified Scatterplots***. Computer Graphics Forum, 2025.

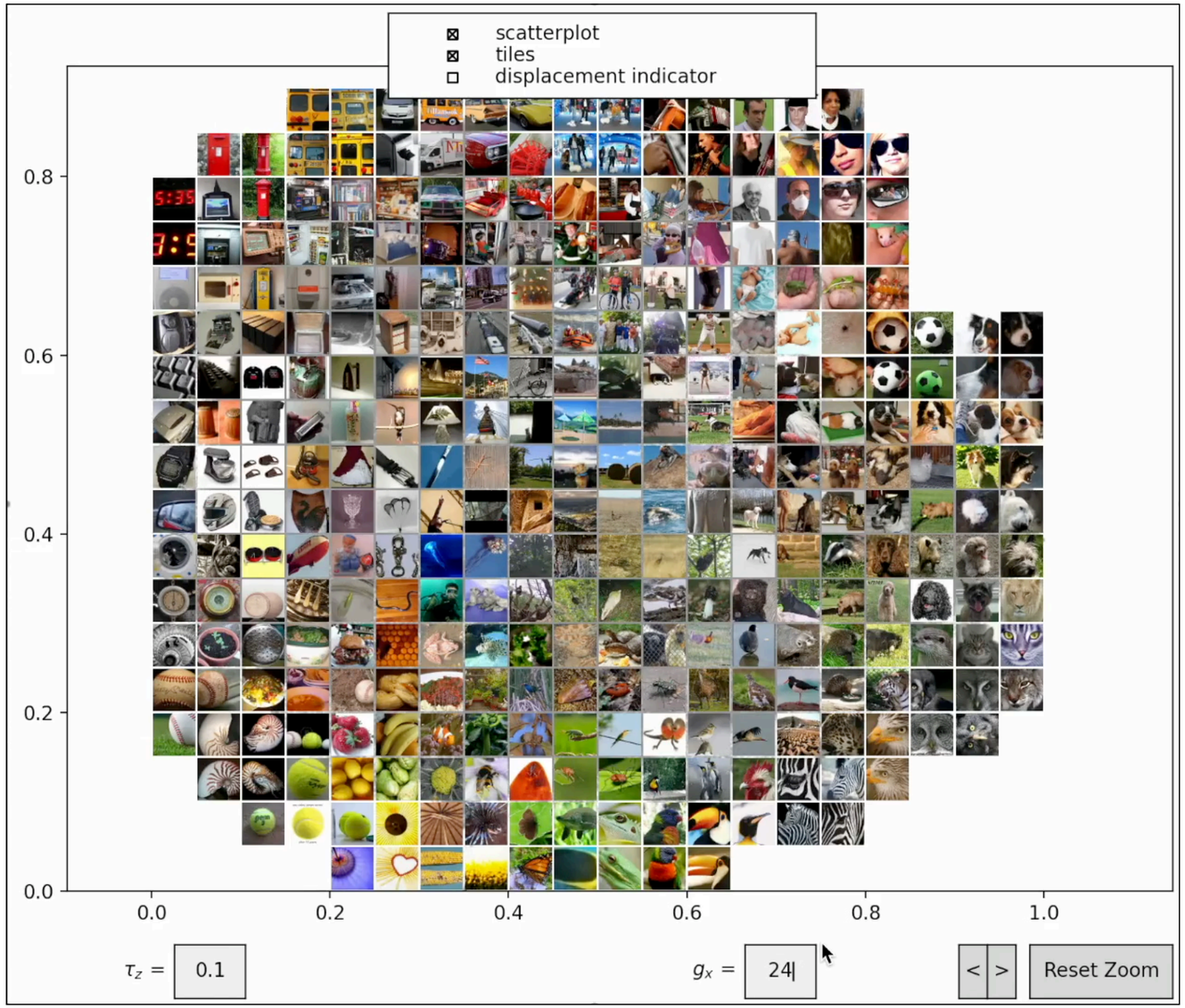
Scaling Study





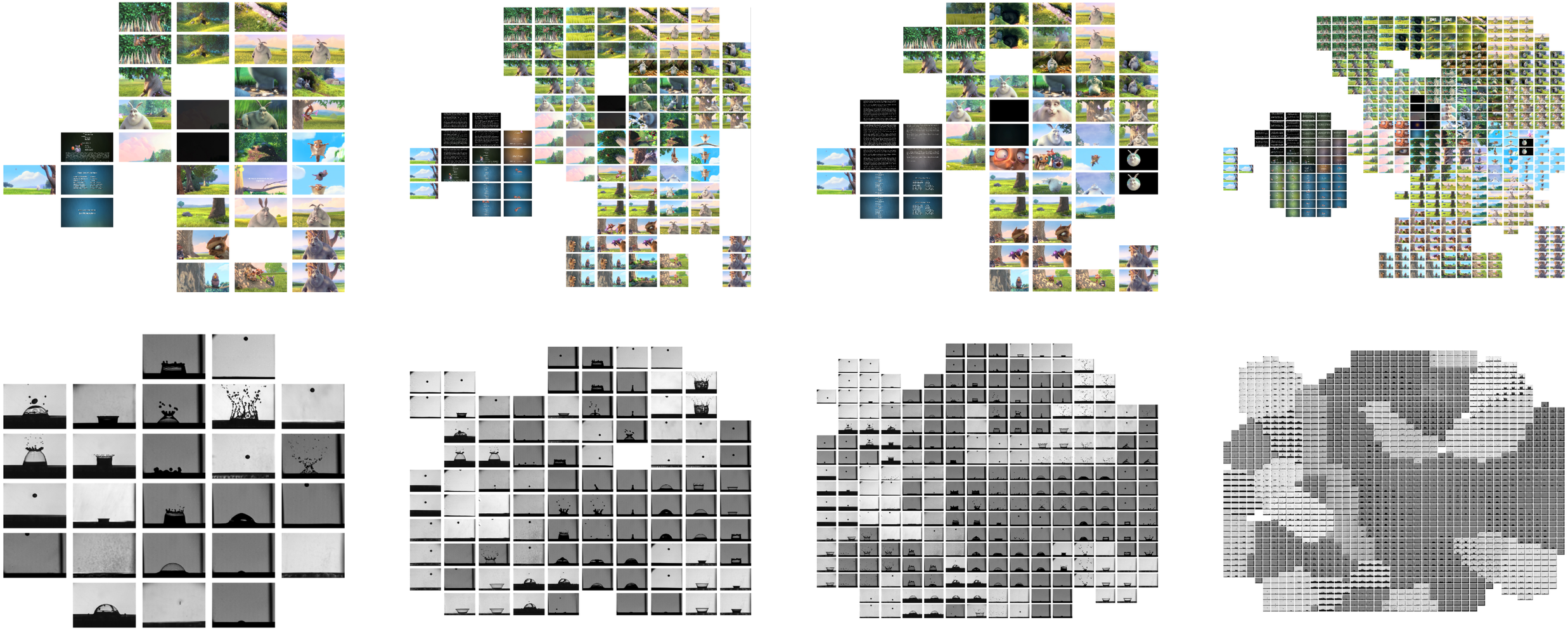
run on Mac Mini M4; synthetic data generated akin to Hilasaca et al. [HMJE∗24]

Case Study: ImageNet (~1.2M points)



https://github.com/freysn/sca2gri

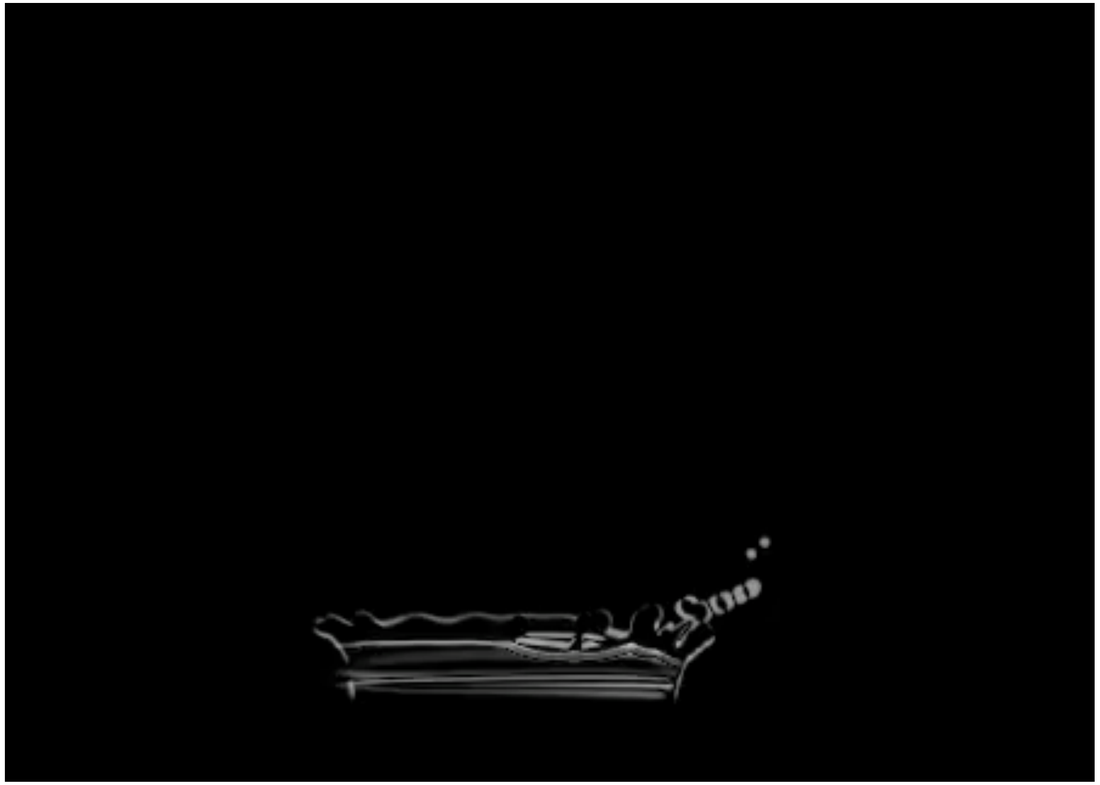
Grids optimized, what about image thumbnails?



⚡⚡⚡increasingly hard to see thumbnail content⚡⚡⚡

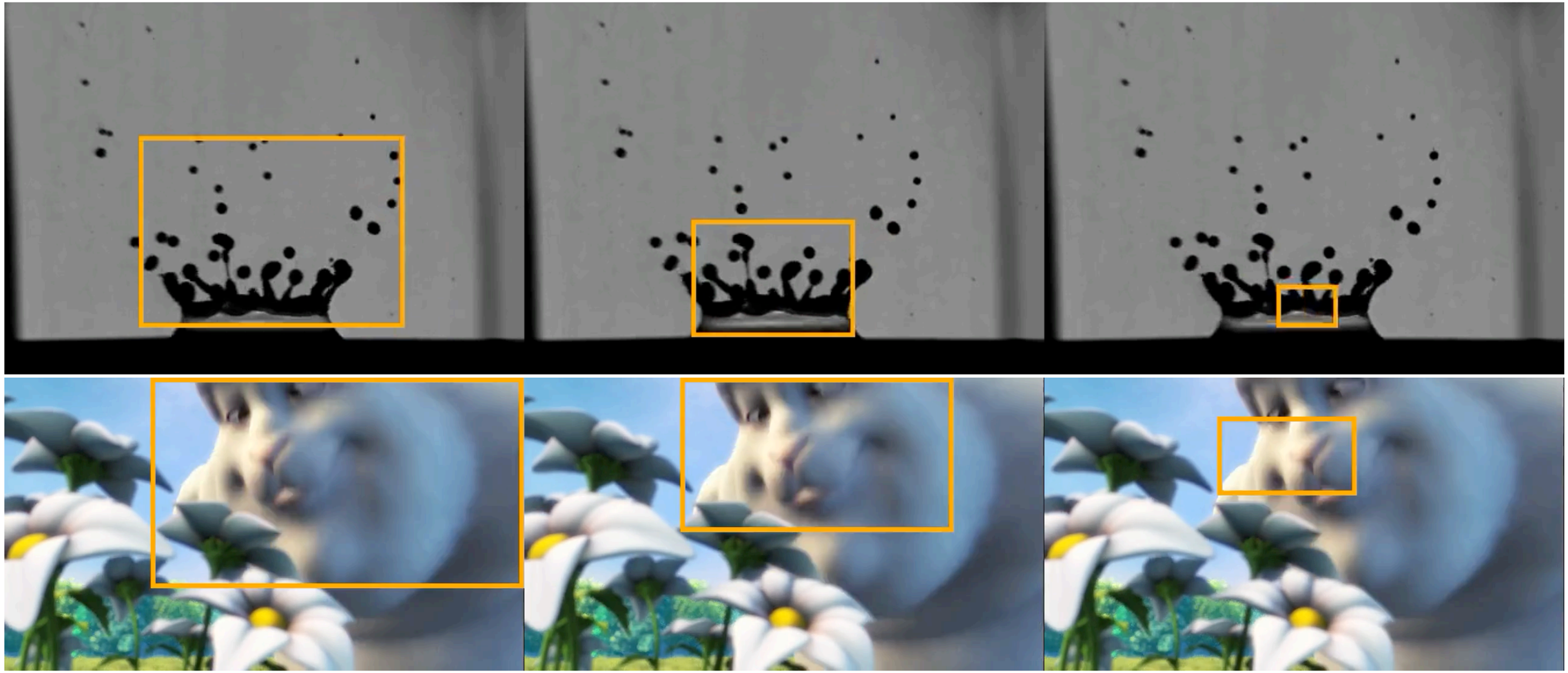
S. Frey, “AGri: Adaptive Thumbnails For Grid-based Visualizations,IEEE VIS (Short), 2025.

identify salient crop areas



S. Frey, “AGri: Adaptive Thumbnails For Grid-based Visualizations,IEEE VIS (Short), 2025.

identify crop areas (maximize salience)



crop sizes — ***large*** ***medium*** ***small***

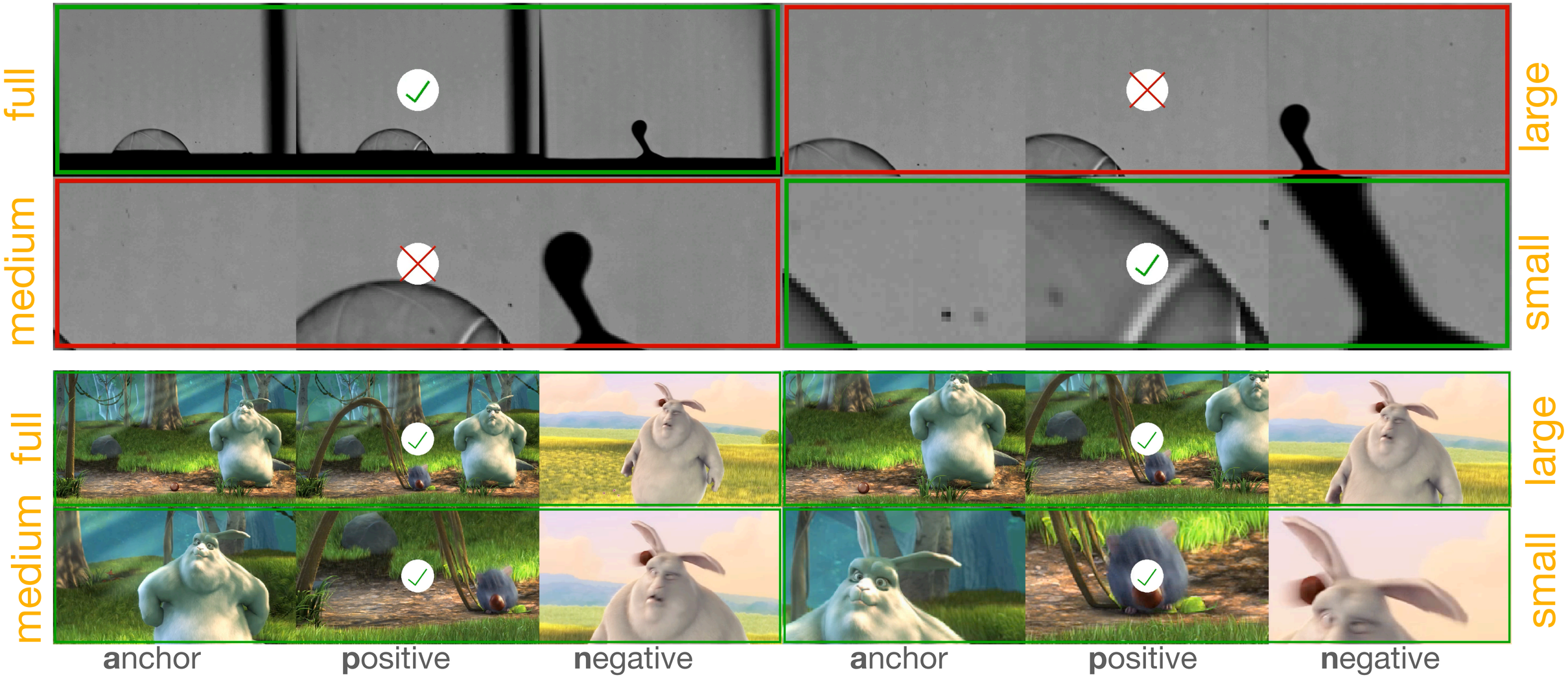
S. Frey, “AGri: Adaptive Thumbnails For Grid-based Visualizations,IEEE VIS (Short), 2025.

Evaluate Expressiveness - Create Triplets

triplet generation: $\Delta_{\text{MSE}}(x_a, x_p) < \Delta_{\text{MSE}}(x_a, x_n)$

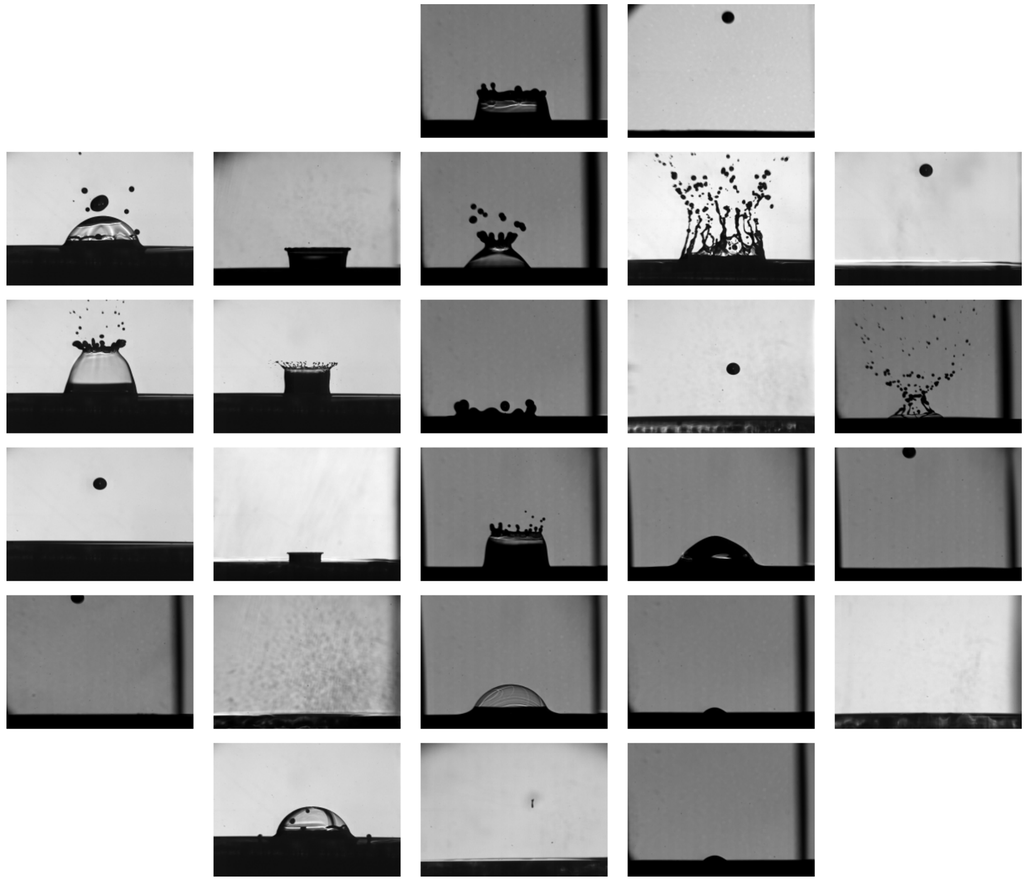
F. Schroff, D. Kalenichenko, and J. Philbin. **Facenet: A unified embedding for face recognition and clustering**. IEEE CVPR, 2015.

Evaluate Expressiveness - Evaluate Triplets

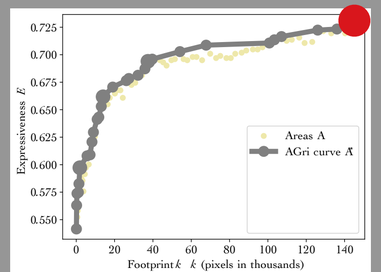


$$E(\mathbf{a}) := \frac{1}{\|T\|} \sum_{(x_a,x_p,x_n)\in T} \Delta_{\text{LPIPS}}\left(\mathbf{a}\left(x_a\right), \mathbf{a}\left(x_p\right)\right) < \Delta_{\text{LPIPS}}\left(\mathbf{a}\left(x_a\right), \mathbf{a}\left(x_n\right)\right)$$

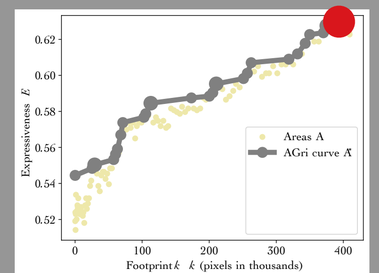
Adaptive Thumbnails – Fixed Screen Resolution (1/4)



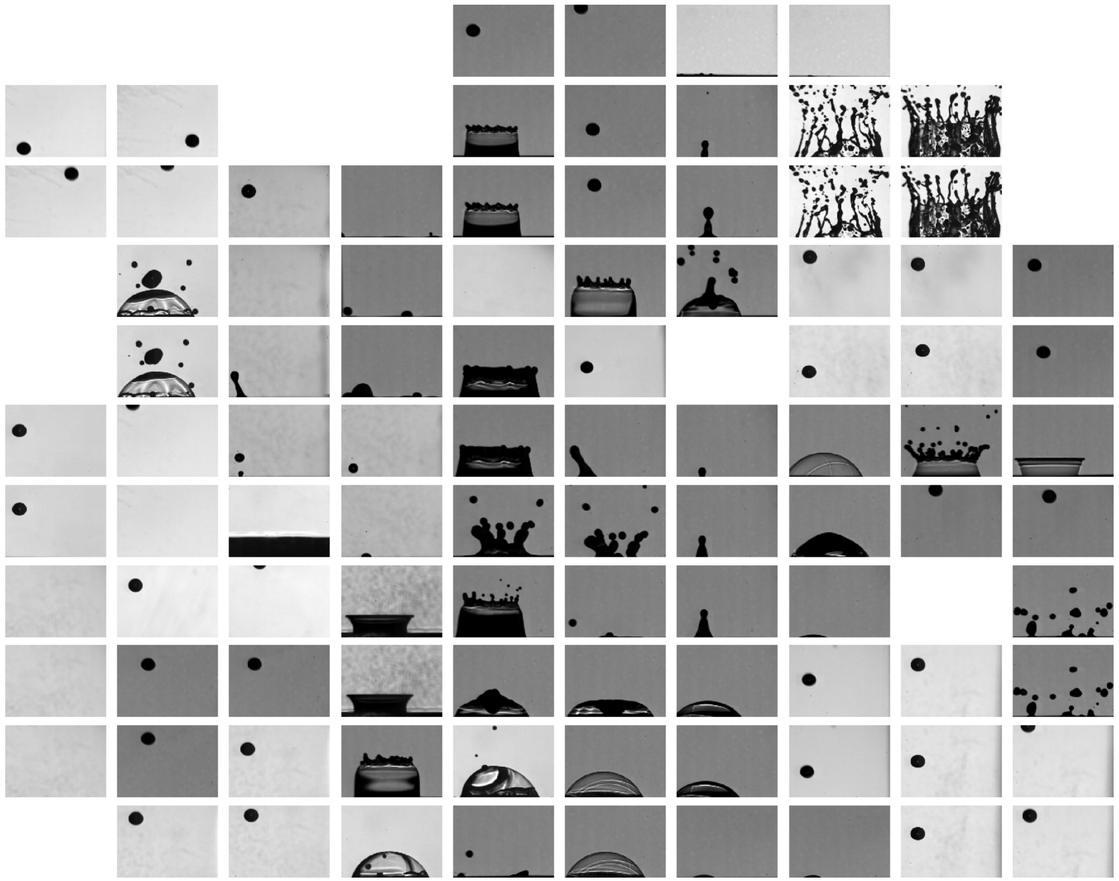
Droplet Splashing Experiments



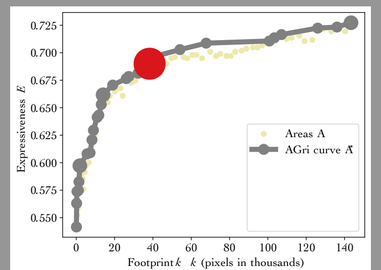
Big Buck Bunny



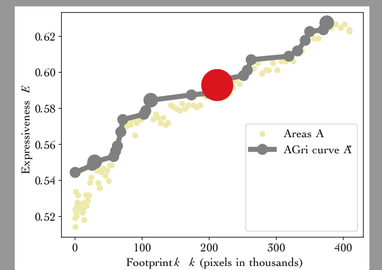
Adaptive Thumbnails – Fixed Screen Resolution (2/4)



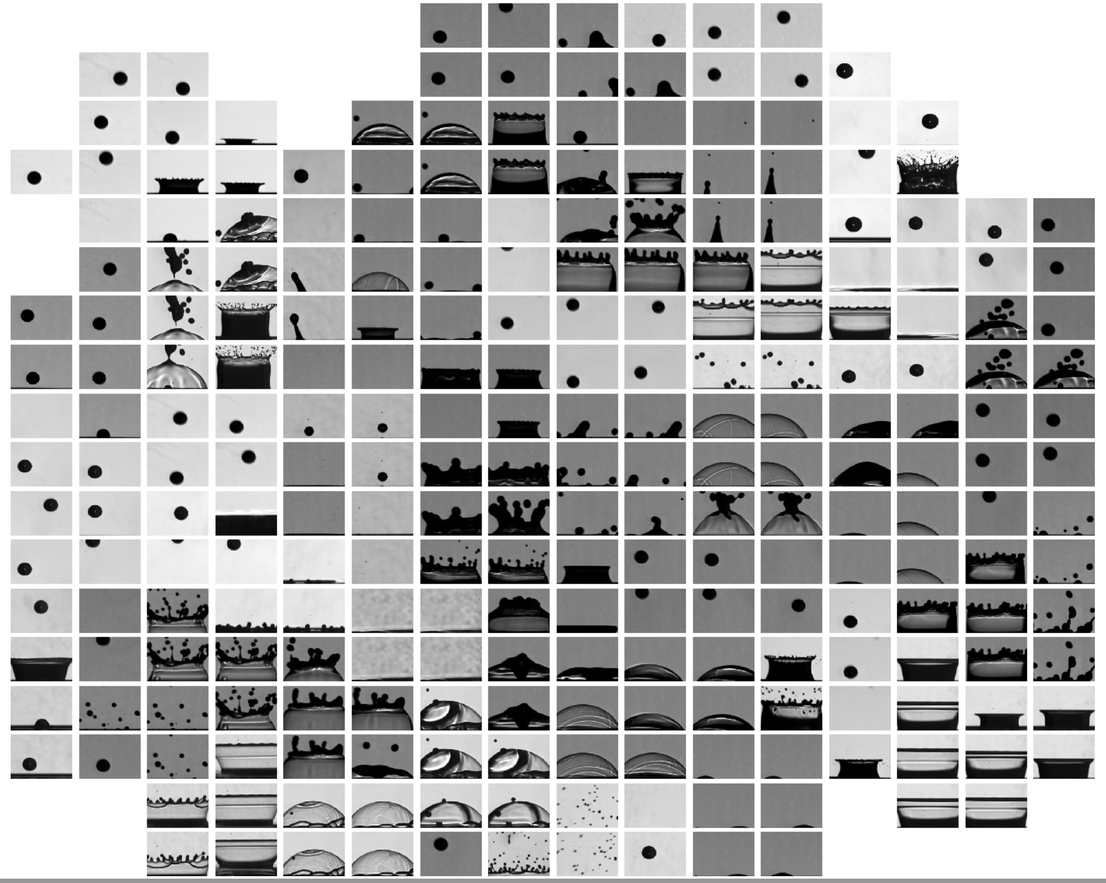
Droplet Splashing Experiments

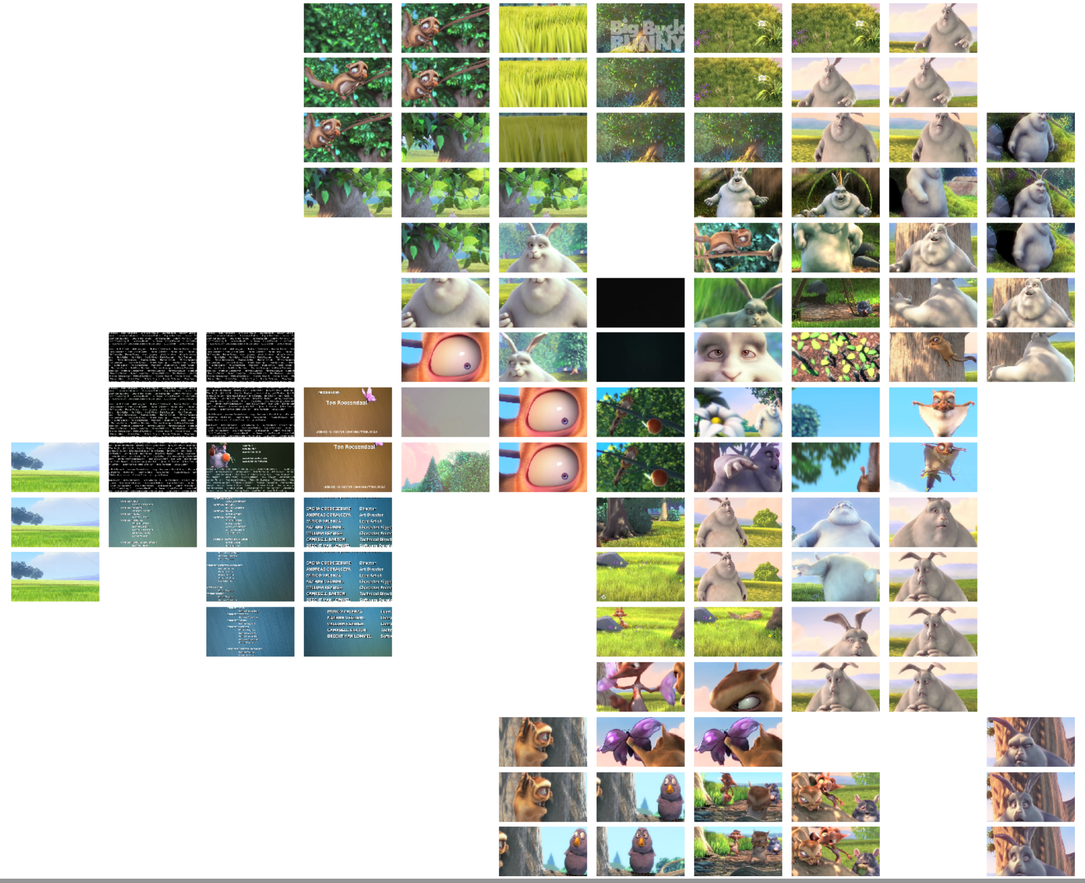


Big Buck Bunny

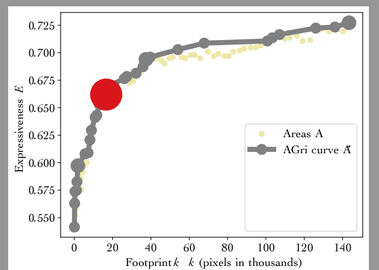


Adaptive Thumbnails – Fixed Screen Resolution (3/4)

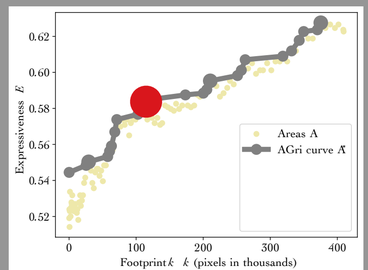




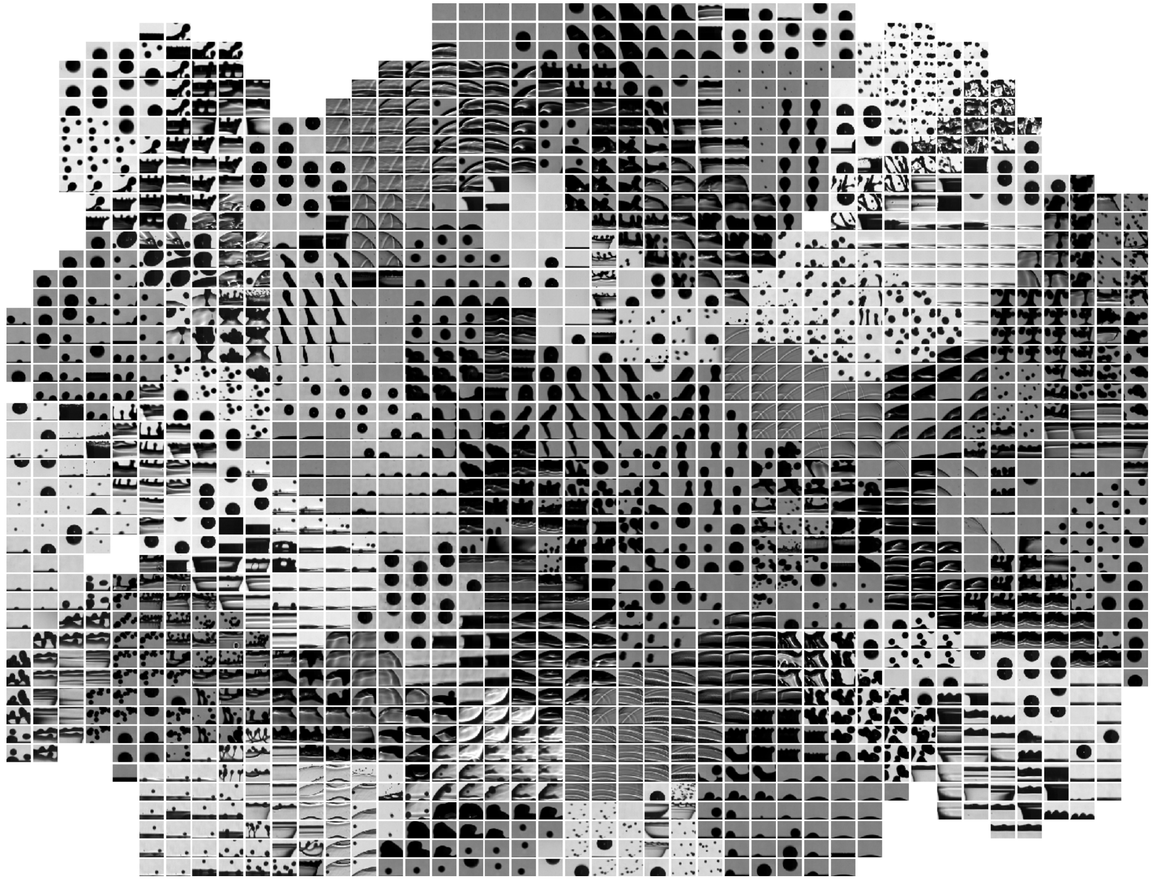
Droplet Splashing Experiments



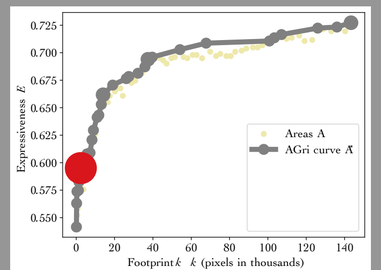
Big Buck Bunny



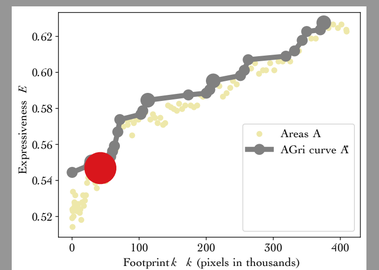
Adaptive Thumbnails – Fixed Screen Resolution (4/4)



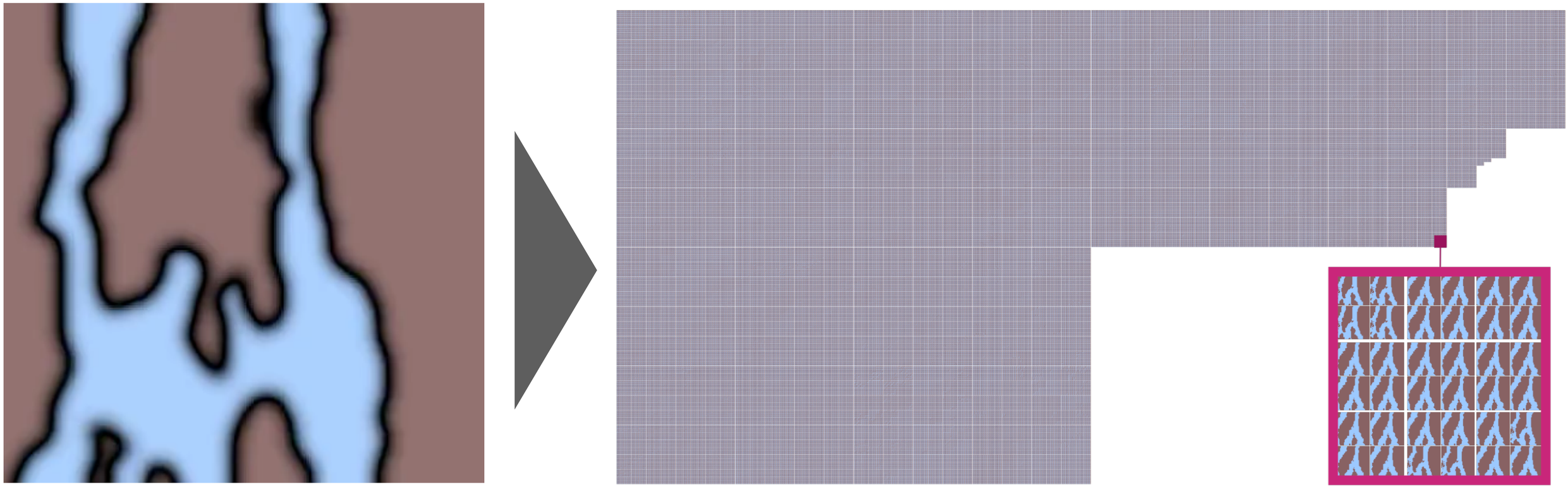
Droplet Splashing Experiments



Big Buck Bunny



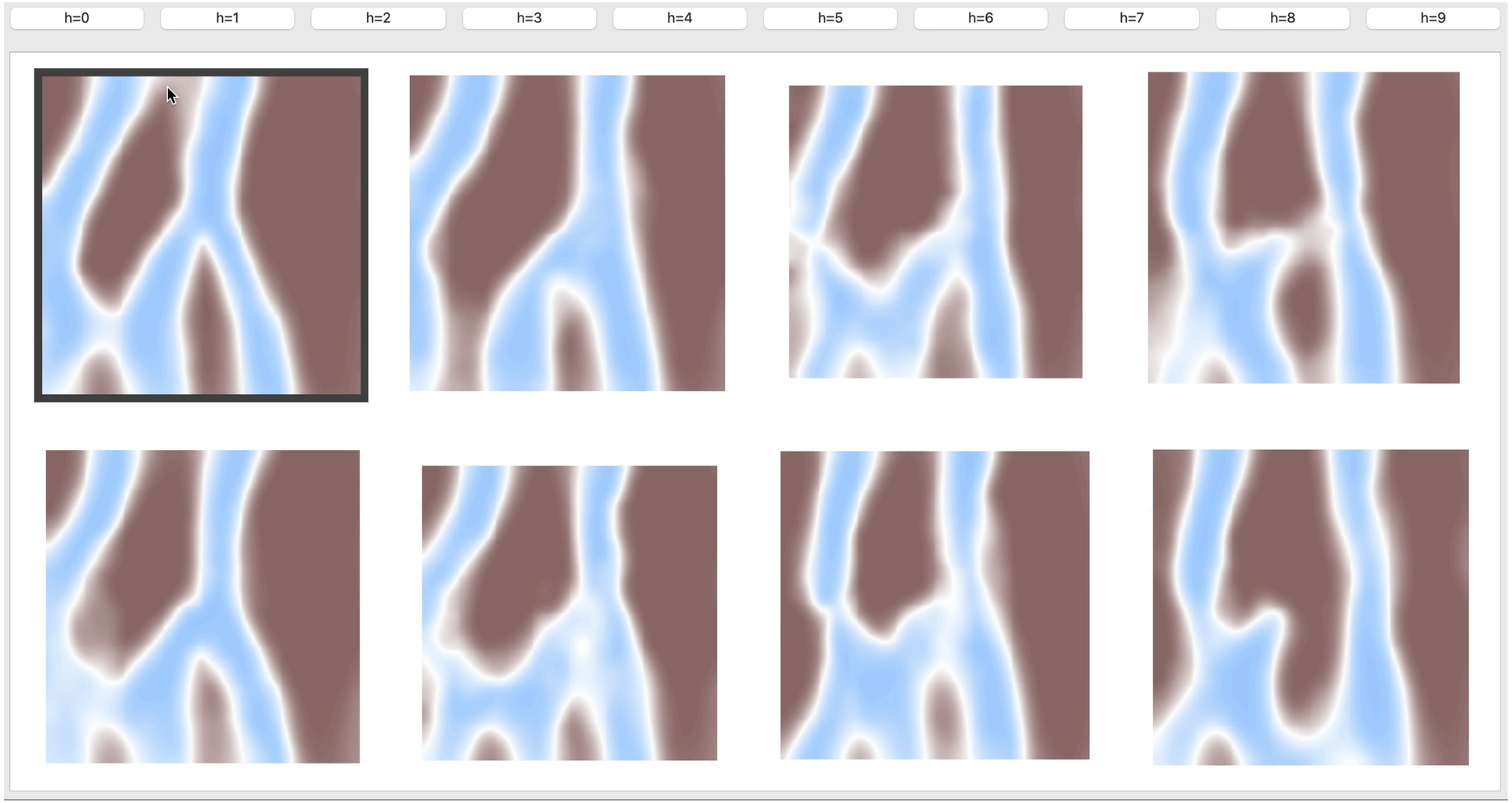
how to represent **ALL** data points?

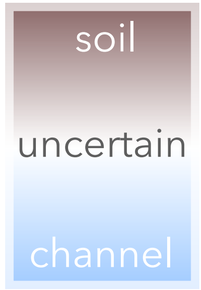


- soil
- channel

S. Reuschen, T. Xu, and W. Nowak, “Bayesian inversion of hierarchical geostatistical models using a parallel-tempering sequential Gibbs MCMC,” Advances in Water Resources, 2020.

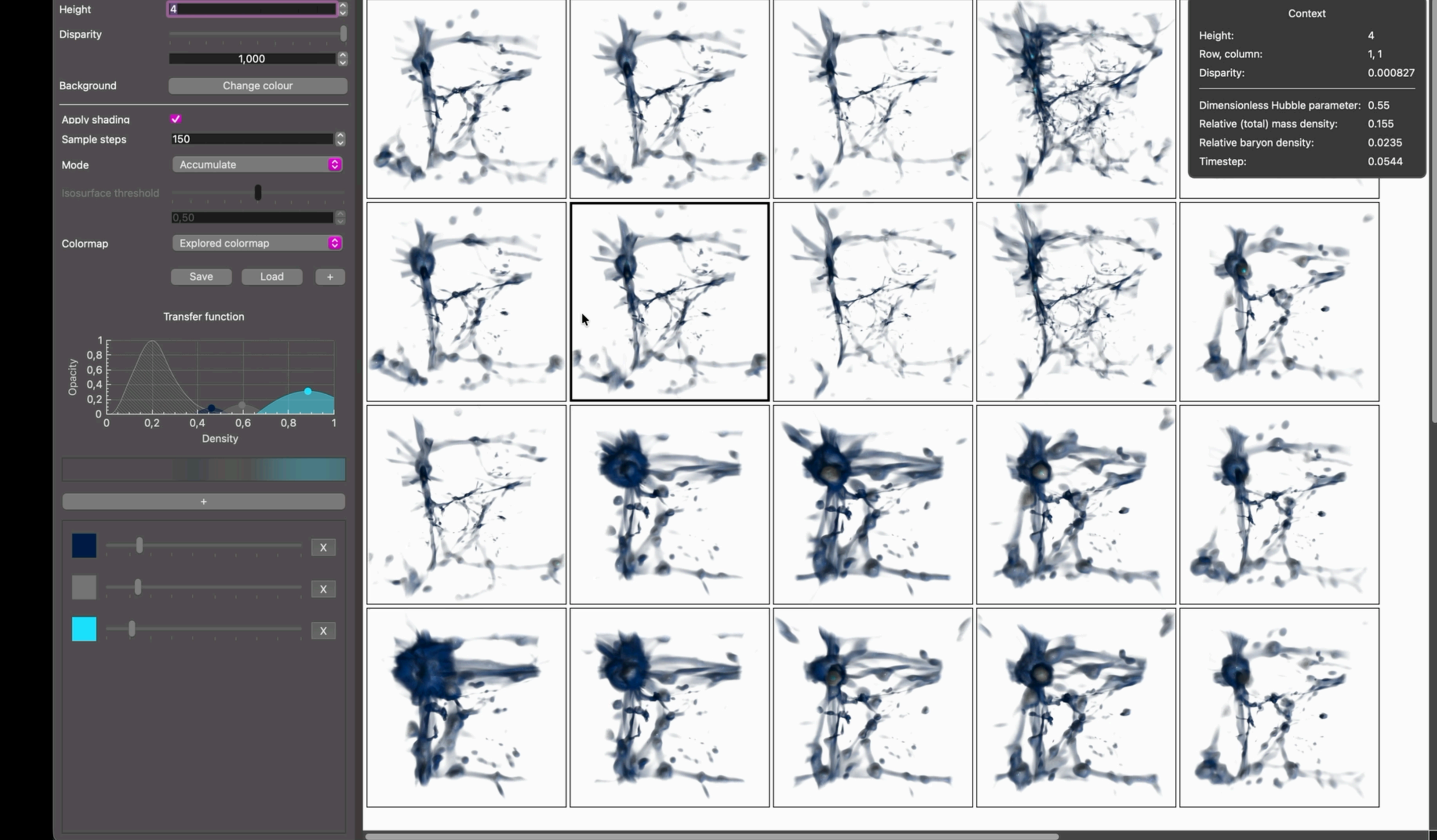
hierarchical grids





Frey, S. (2022). Optimizing Grid Layouts for Level-of-Detail Exploration of Large Data Collections. Computer Graphics Forum, 41(3), 247–258.

combines juxtaposition and explicit encoding


Height
4
Disparity
1,000
Background
Change colour
Apply shading
Sample steps
150
Mode
Accumulate
Isosurface threshold
0,50
Colormap
Explored colormap
Save
Load
+
Transfer function
Opacity
Density
0
0,2
0,4
0,6
0,8
1
X
Context
Height: 4
Row, column: 1, 1
Disparity: 0.000827
Dimensionless Hubble parameter: 0.55
Relative (total) mass density: 0.155
Relative baryon density: 0.0235
Timestep: 0.0544

Hieradex

Disparity

1,000

Background Change colour

Volume rendering

General

Apply shading

Sample steps 200

Mode Accumulate

Isosurface threshold 0,50

Memory

Available memory 1,00

Average LOD: 6.07 **Average SSIM:** 0.99

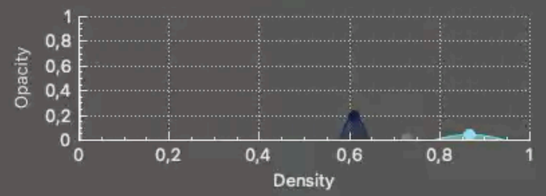
Show LOD overlay

0 1 2 3 4 5 6 7

Transfer function

Colormap Explored colormap

Save Load +



+

X

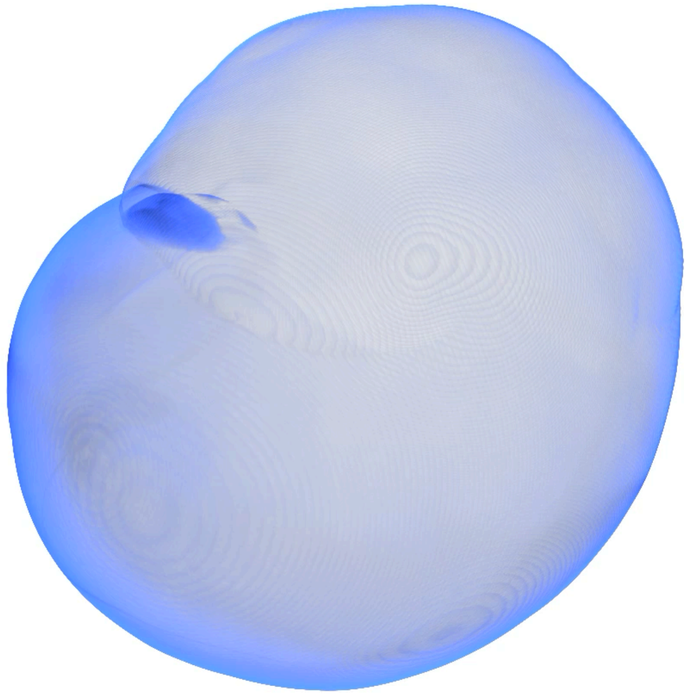
X

X

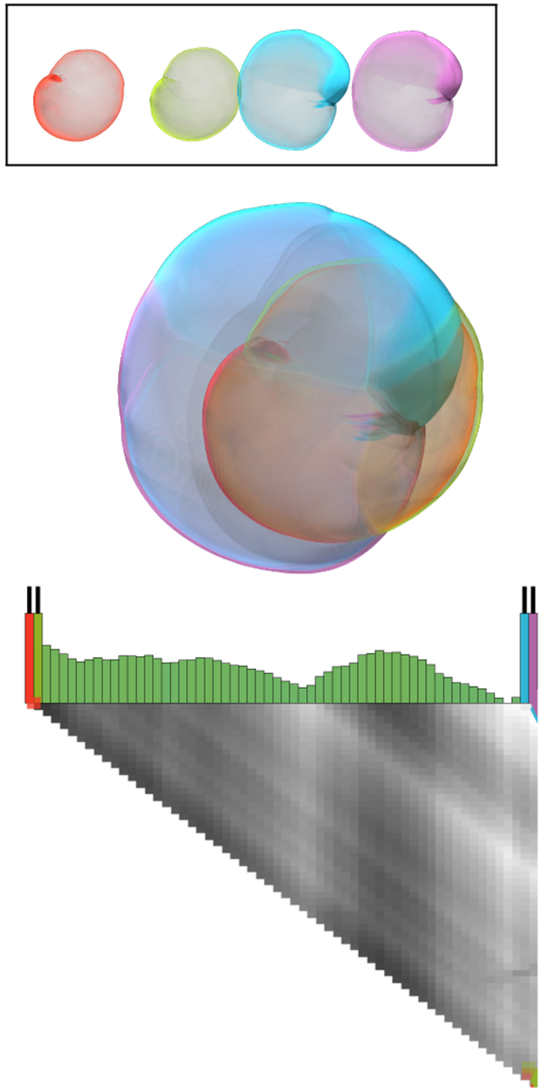
How to quantify similarity between data points?

Crucial for many large data visualization techniques!

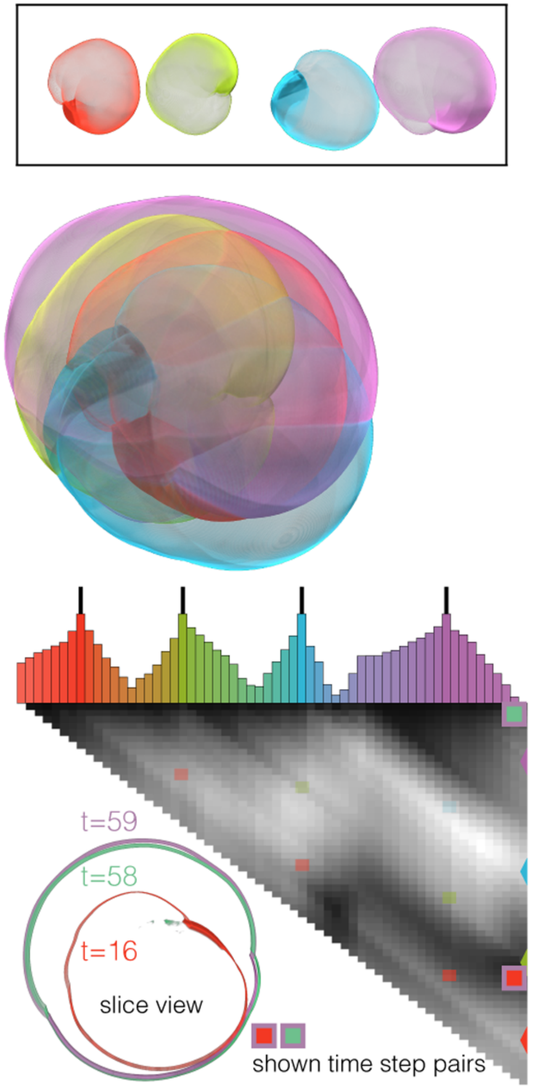
00000



element-based
(MSE, PSNR, ...)



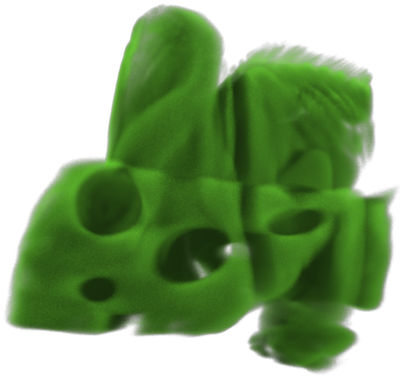
flow-based
(EMD / Wasserstein)



Flow-based Distances

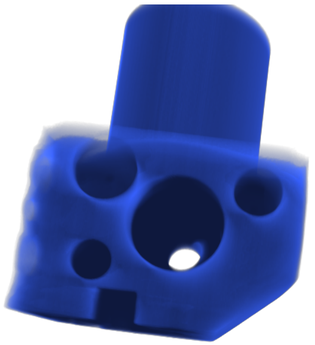
0%

30%



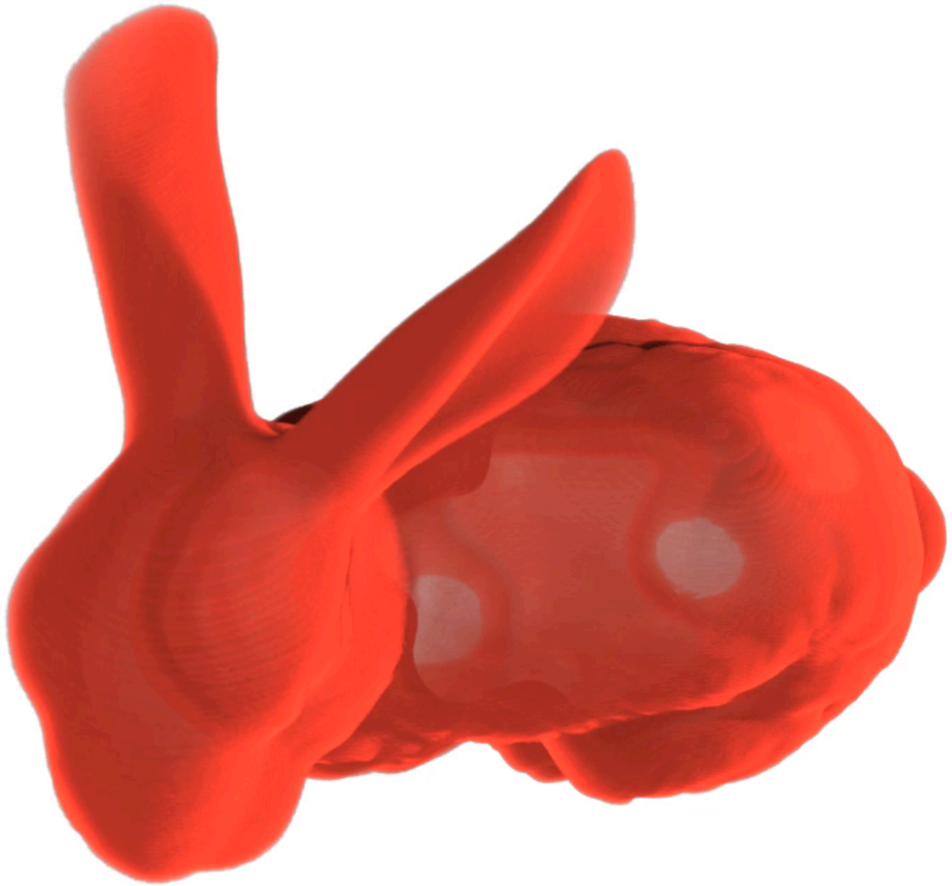
50%

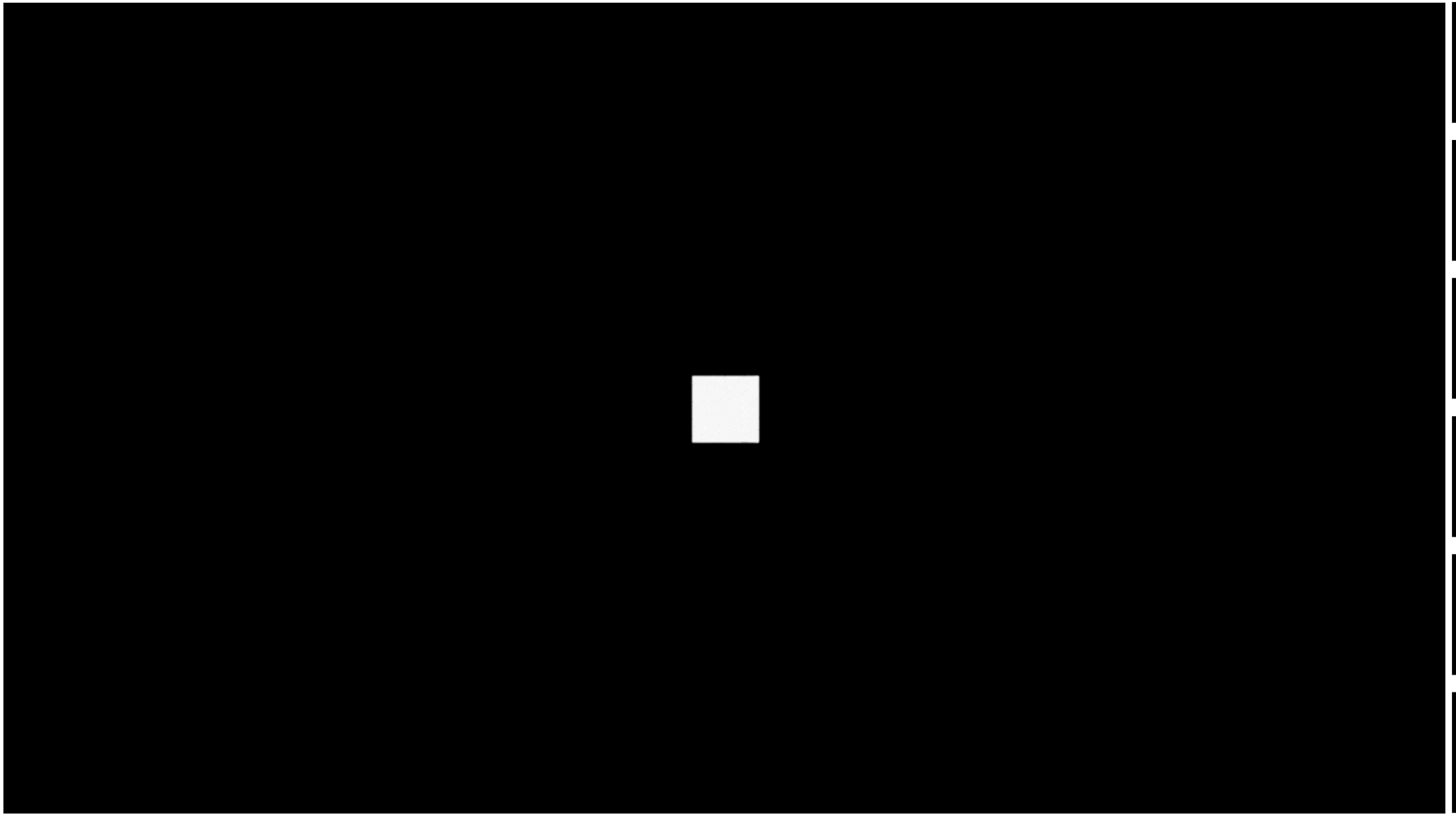
70%

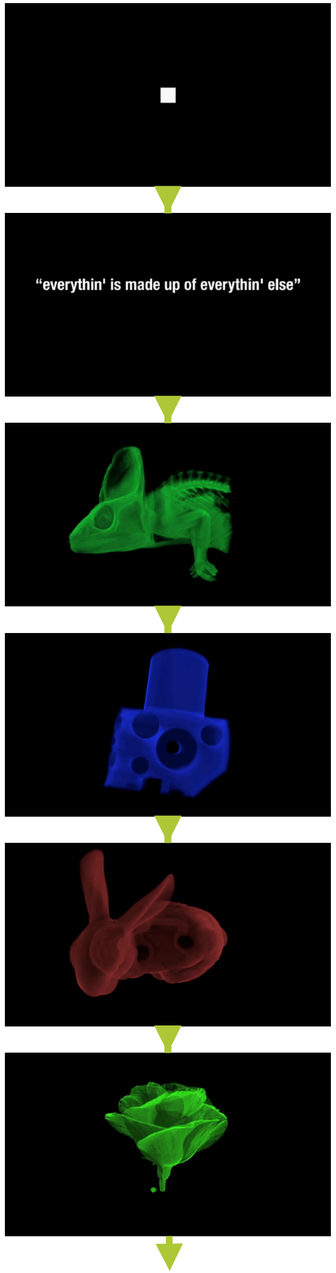


100%

Flow-based Distances

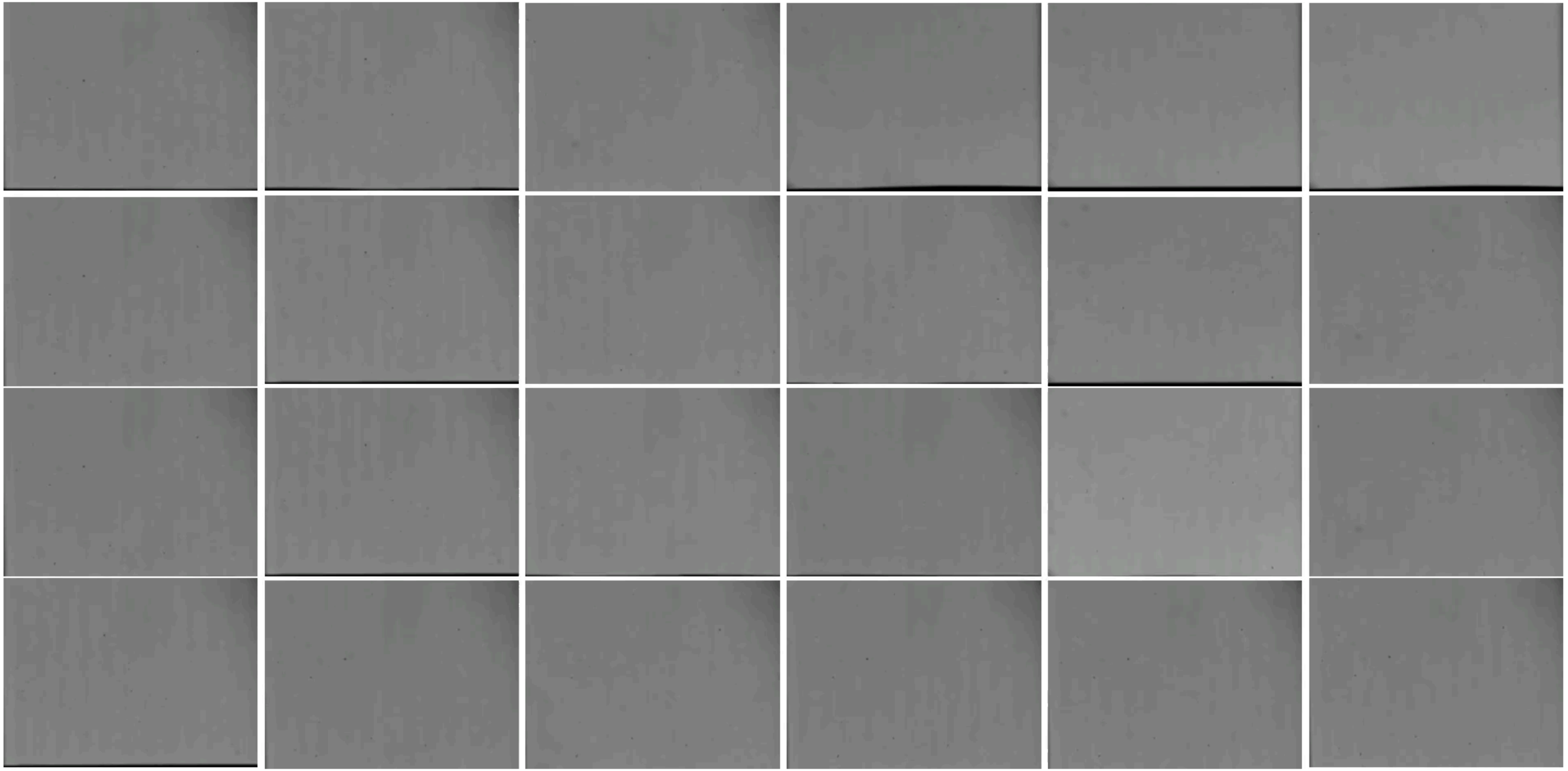


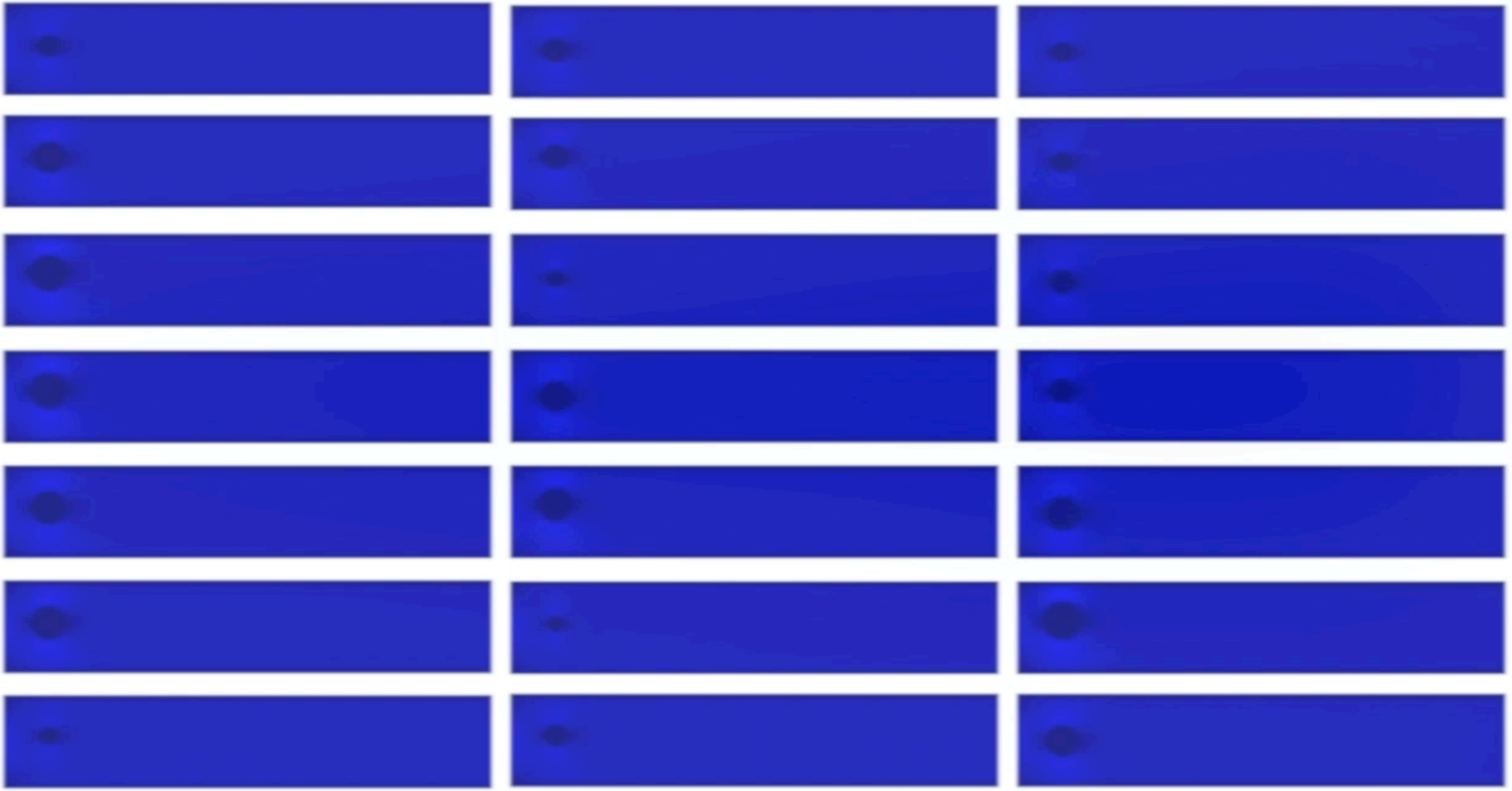




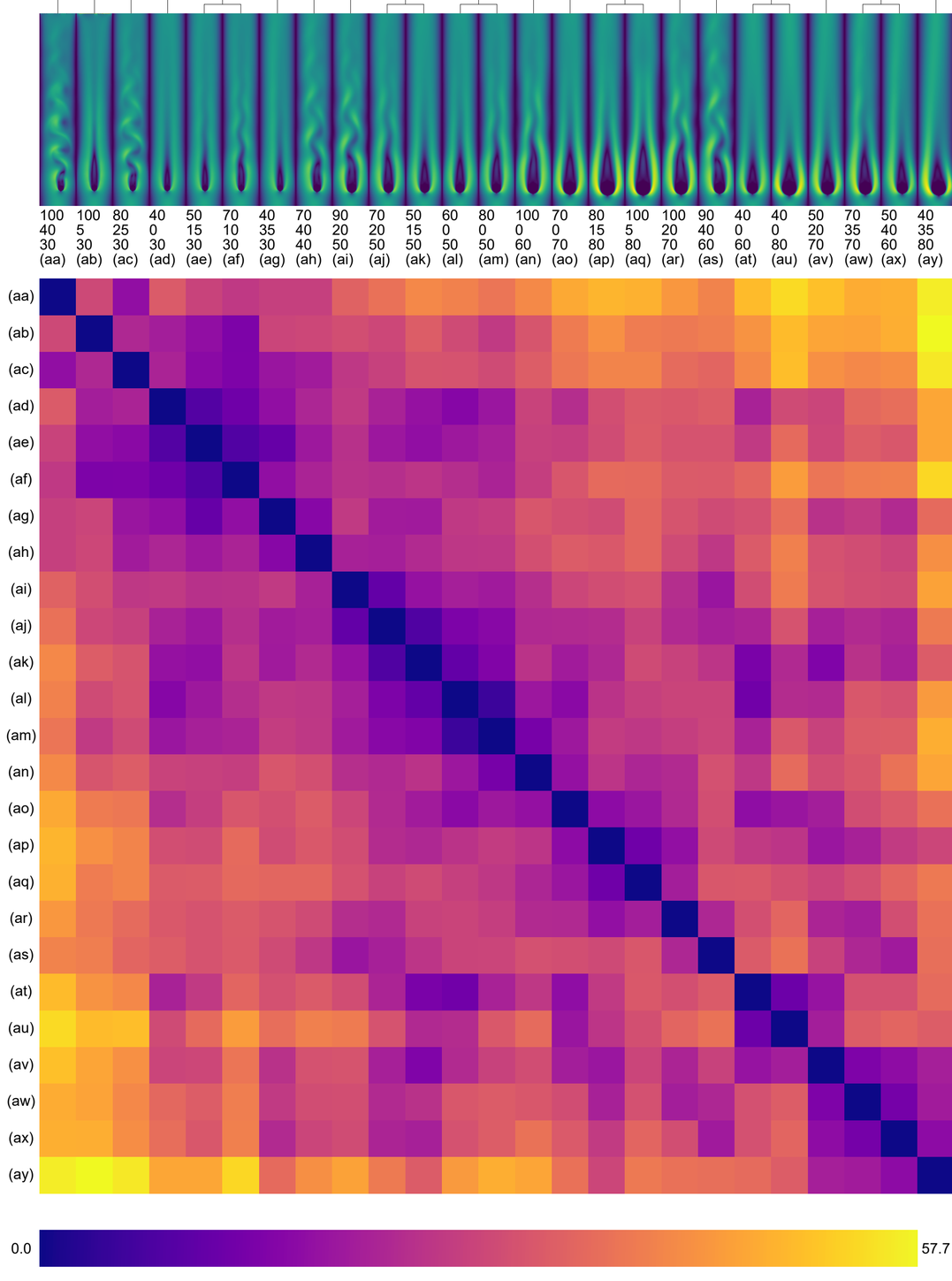
S. Frey and T. Ertl. "Progressive Direct Volume-to-Volume Transformation." *IEEE Transactions on Visualization and Computer Graphics* 23, 2017

…





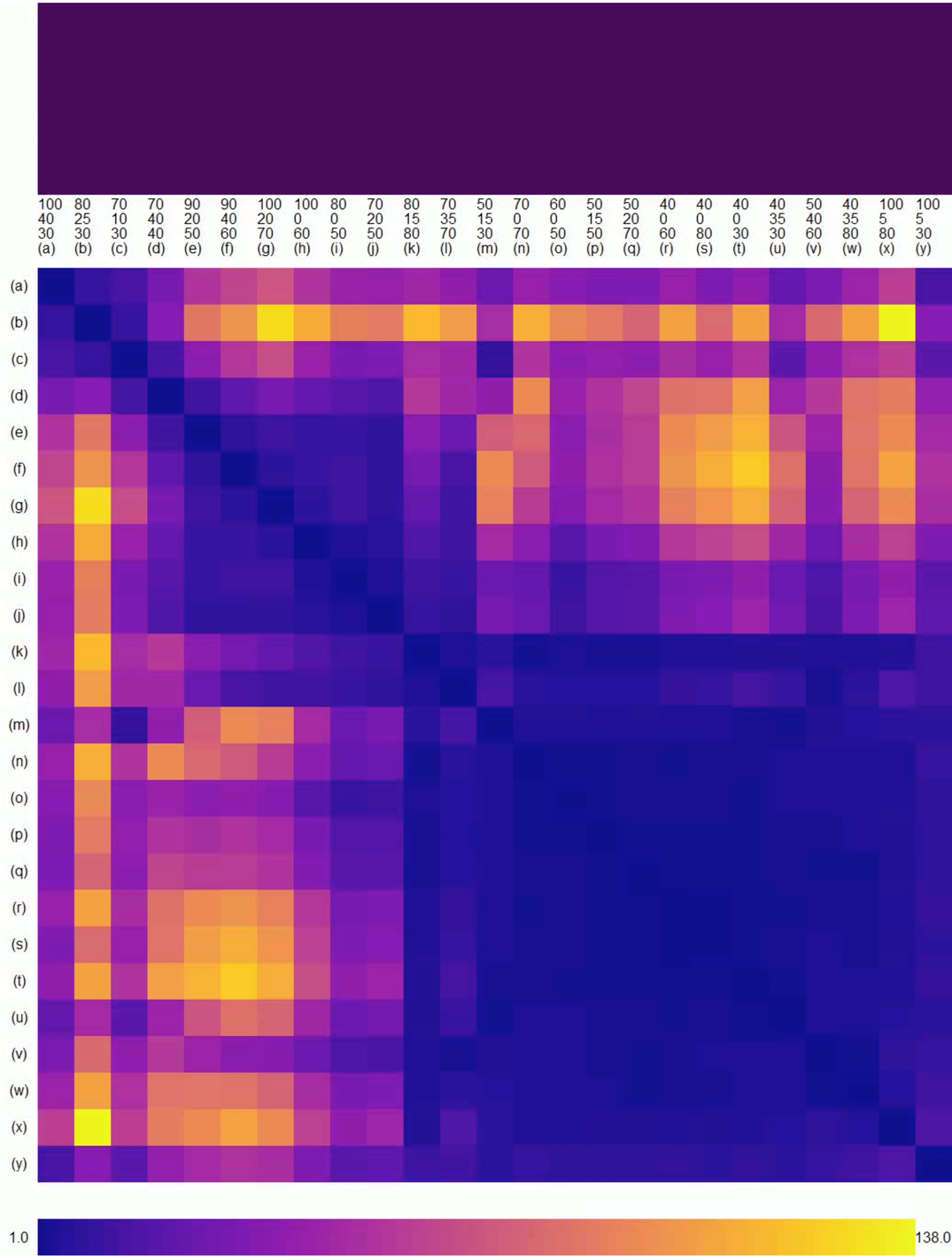
Limited Expressiveness of Spatial Distances



G. Tkachev, S. Frey, and T. Ertl, “Local Prediction Models for Spatiotemporal Volume Visualization,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 27, no. 7, pp. 3091–3108, Jul. 2021.

Limited Expressiveness of Spatial Distances

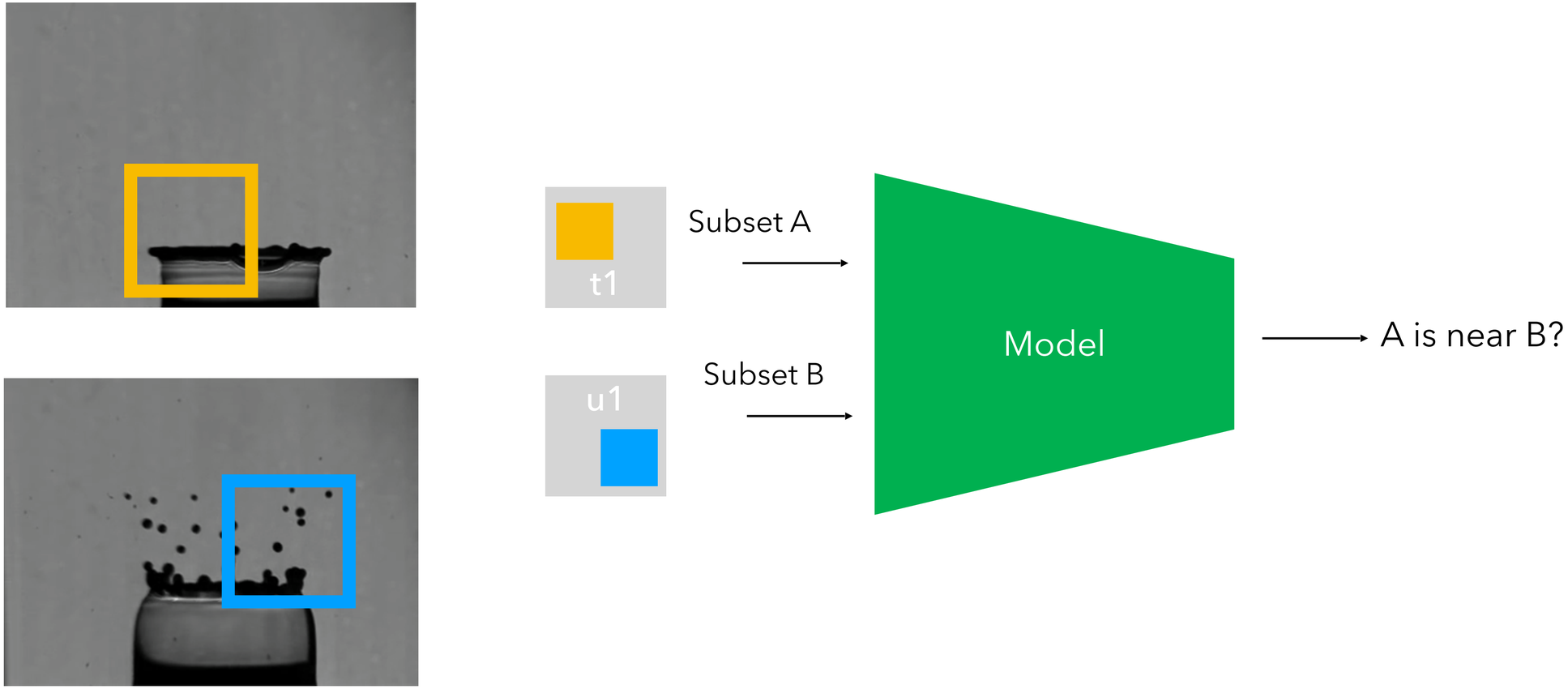
→ self-supervised feature learning



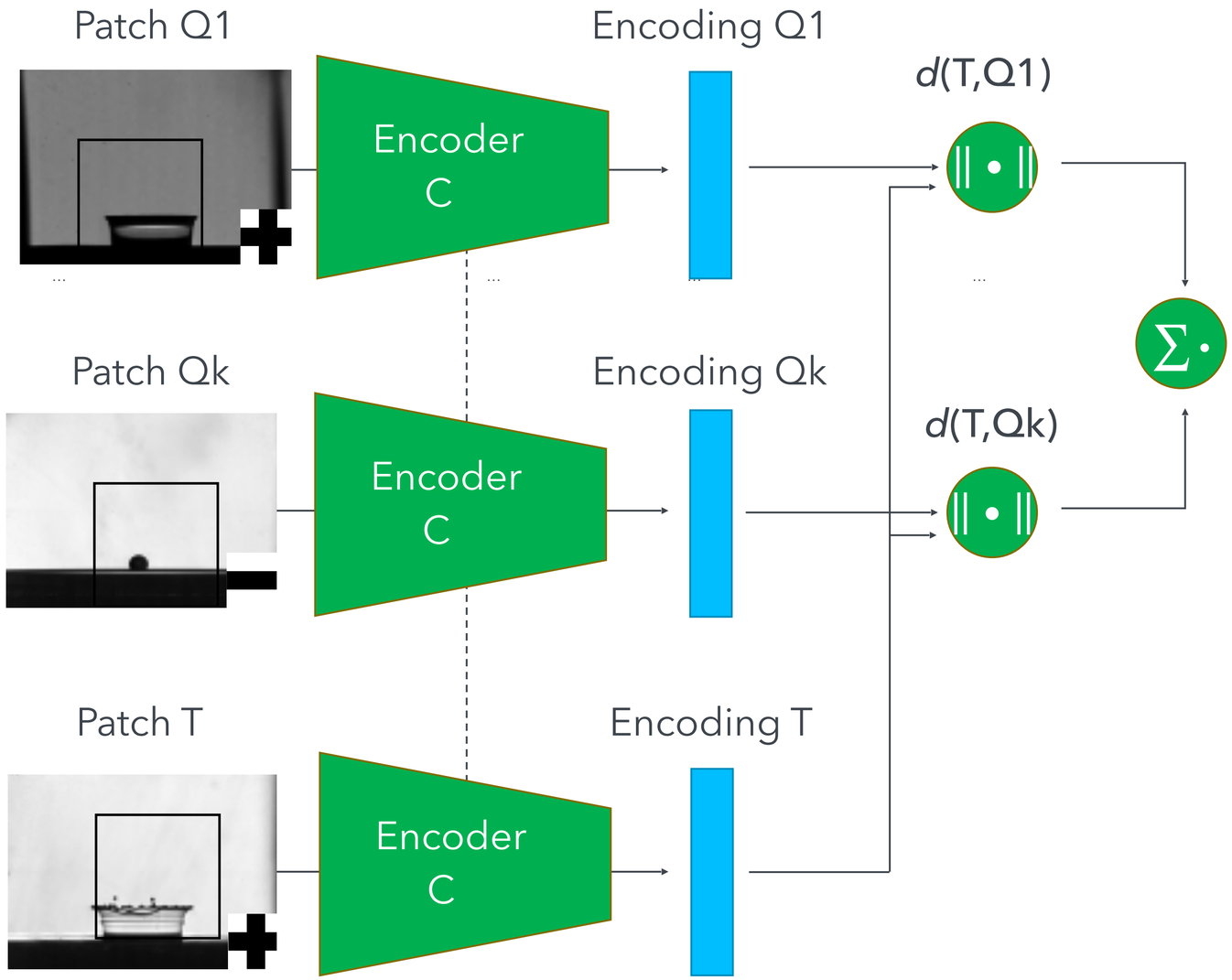
G. Tkachev, S. Frey, and T. Ertl, "Local Prediction Models for Spatiotemporal Volume Visualization," *IEEE Transactions on Visualization and Computer Graphics*, vol. 27, no. 7, pp. 3091–3108, Jul. 2021.

Self-Supervised Learning

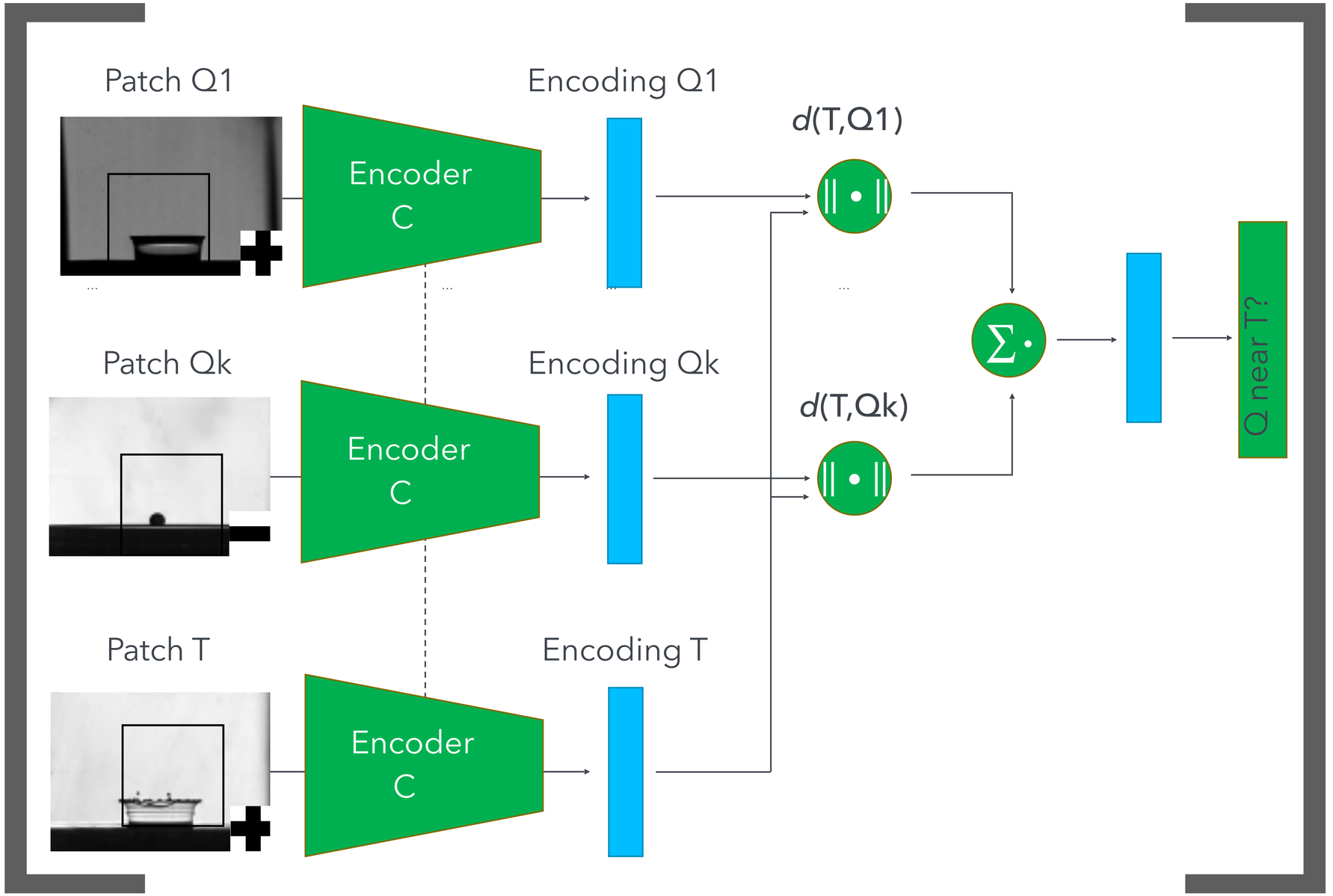
surrogate task: (1) interpolate missing frames (2) extract data flow

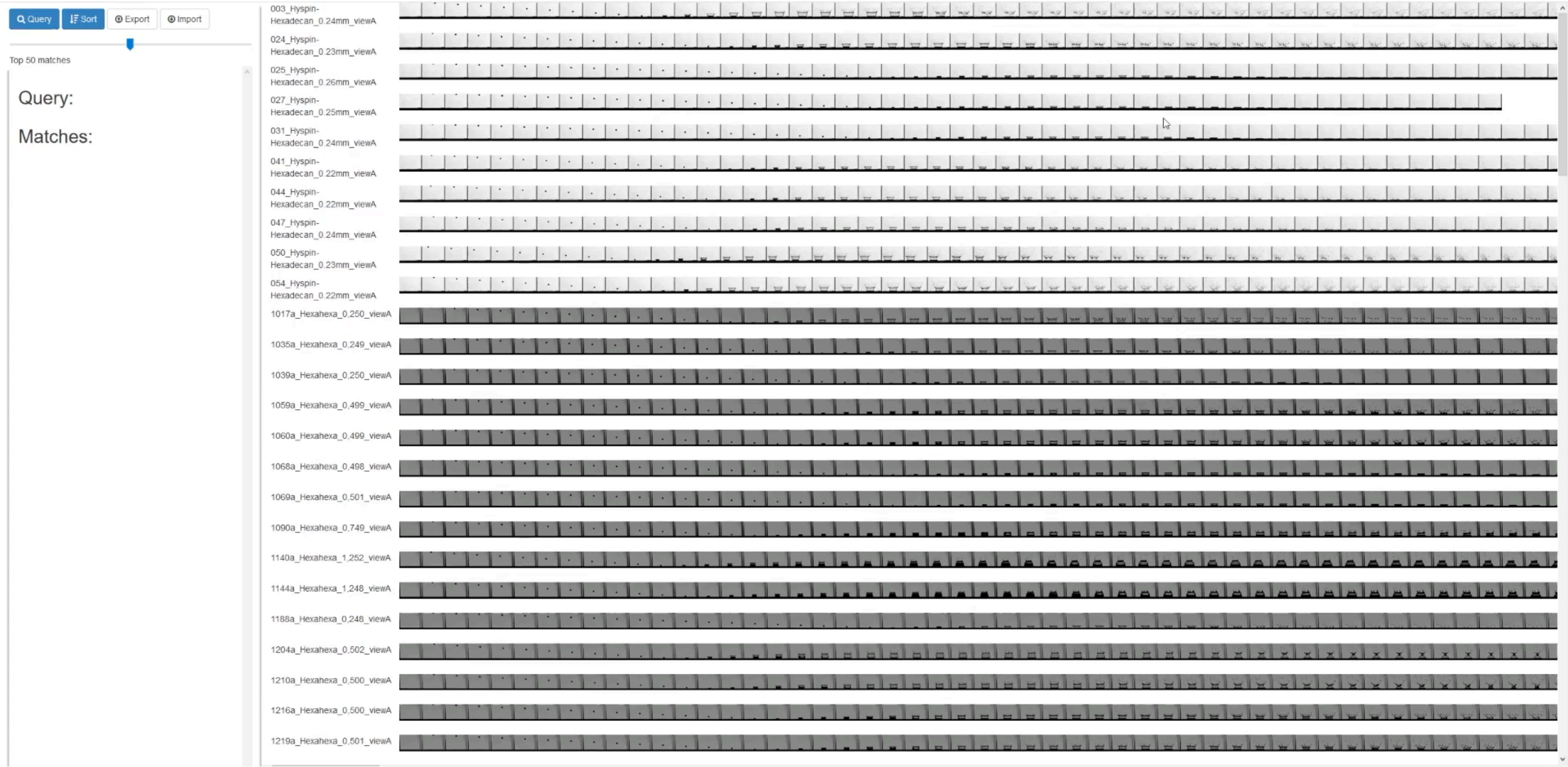


Tkachev, **Frey**, Ertl, S4: Self-Supervised learning of Spatiotemporal Similarity, TVCG, 2021.

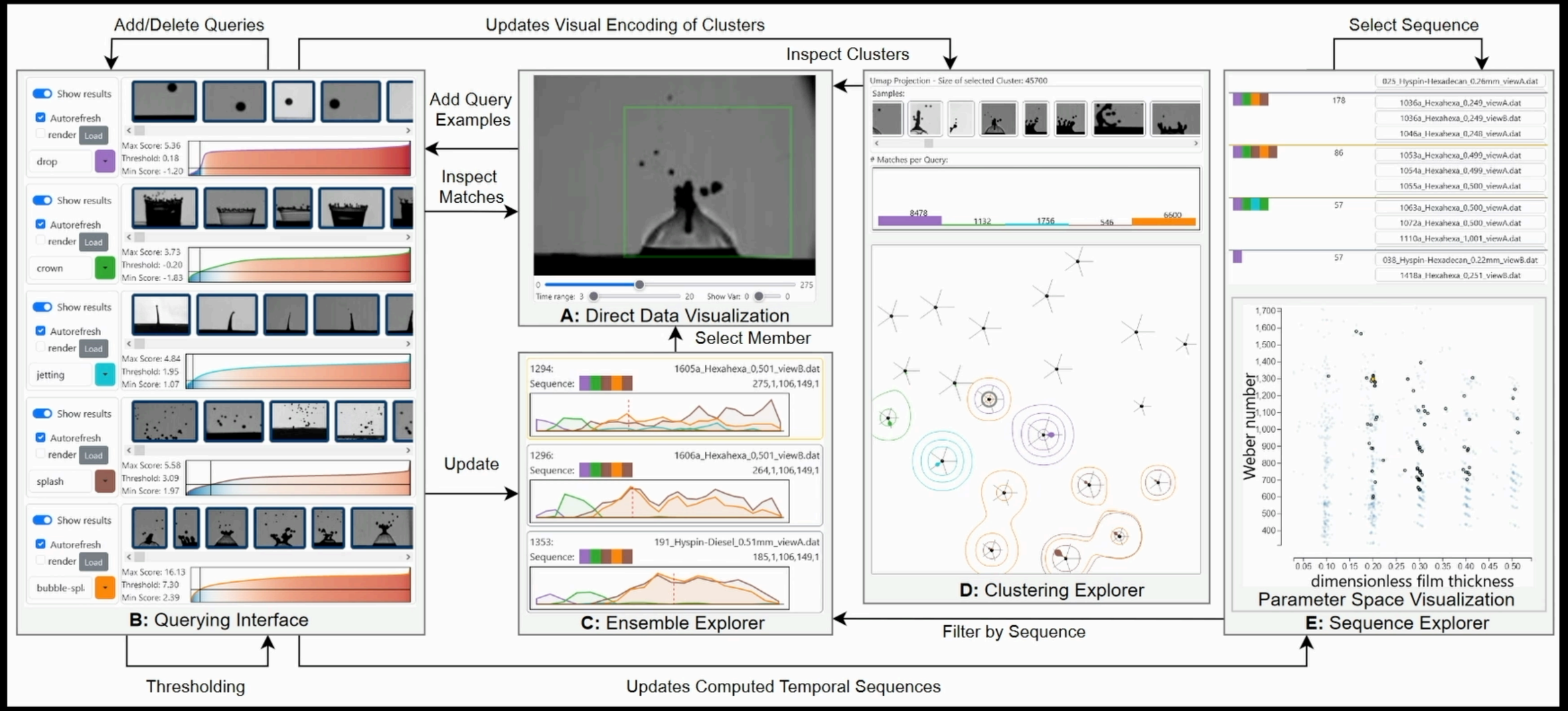

Patch Q1
Encoding Q1
Encoder
C
d(T,Q1)
Patch Qk
Encoding Qk
Encoder
C
d(T,Qk)
Patch T
Encoding T
Encoder
C

$$\min_{C} \mathbb{E}_{(Q,T,Q \text{ near } T)\sim p_{\text{data}}}$$





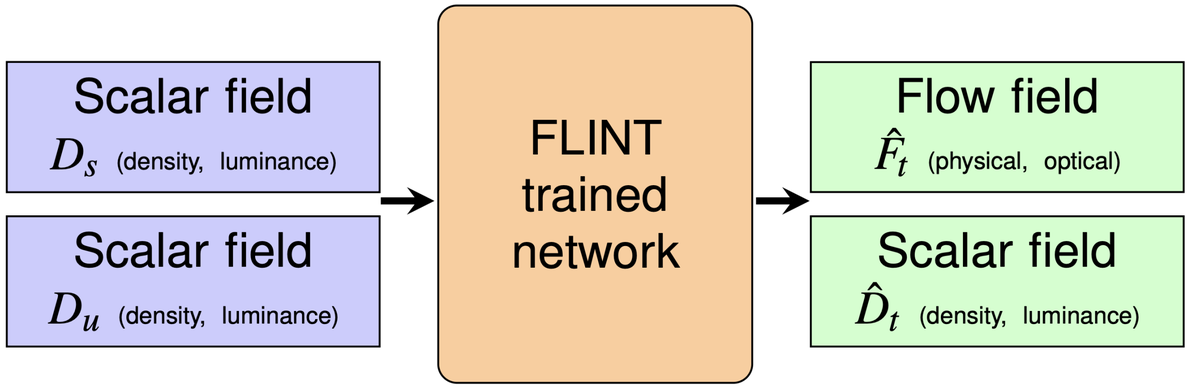
Tkachev, Frey, Ertl, S4: Self-Supervised learning of Spatiotemporal Similarity, TVCG, 2021.



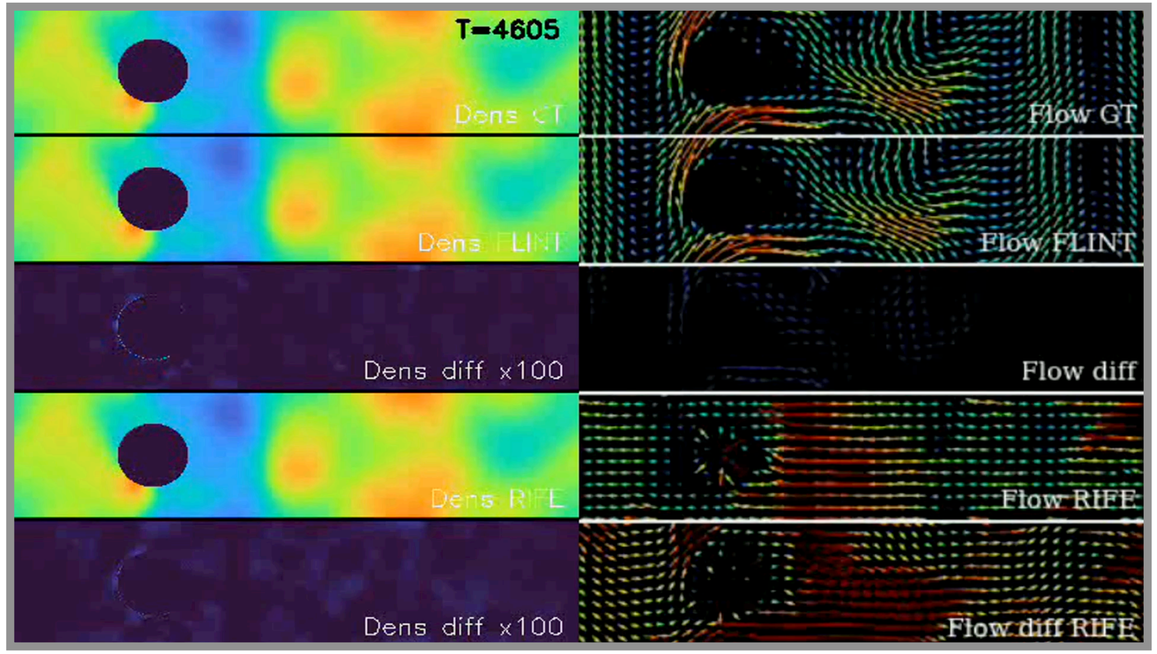
R. Bauer, Q. Ngo, G. Reina, S. Frey, M. Sedlmair, QVis: Query-Based Visual Analysis of Multiscale Patterns in Spatiotemporal Ensembles, TVCG, 2025.

Flint: Temporal Superresolution / Flow

Task: (1) interpolate missing frames (2) extract data flow

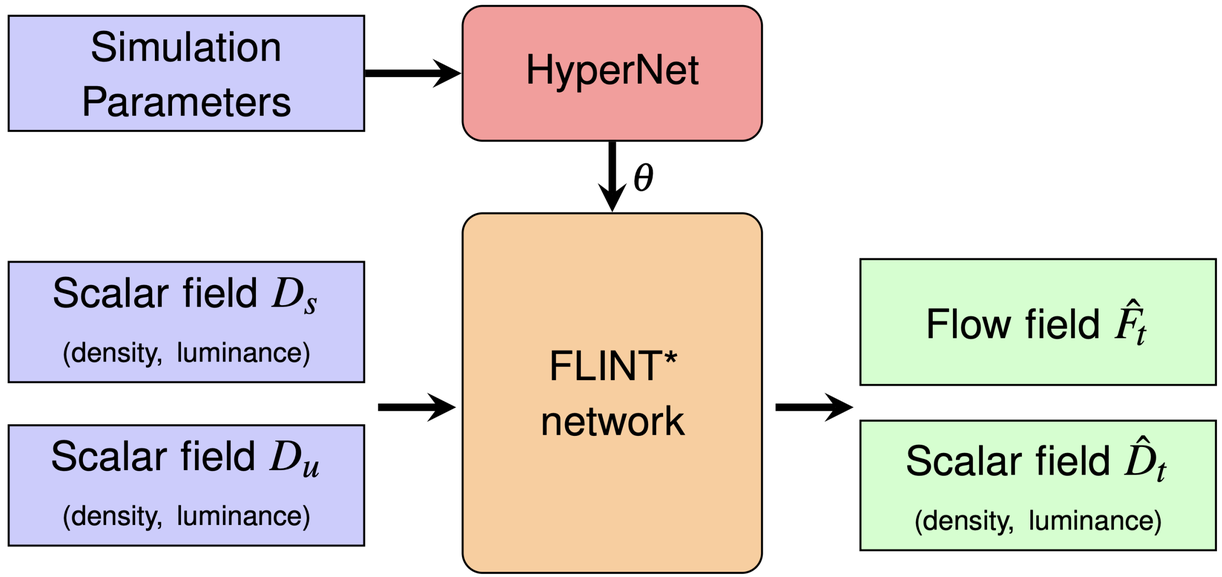


$$D_s < \hat{D}_t < D_u$$

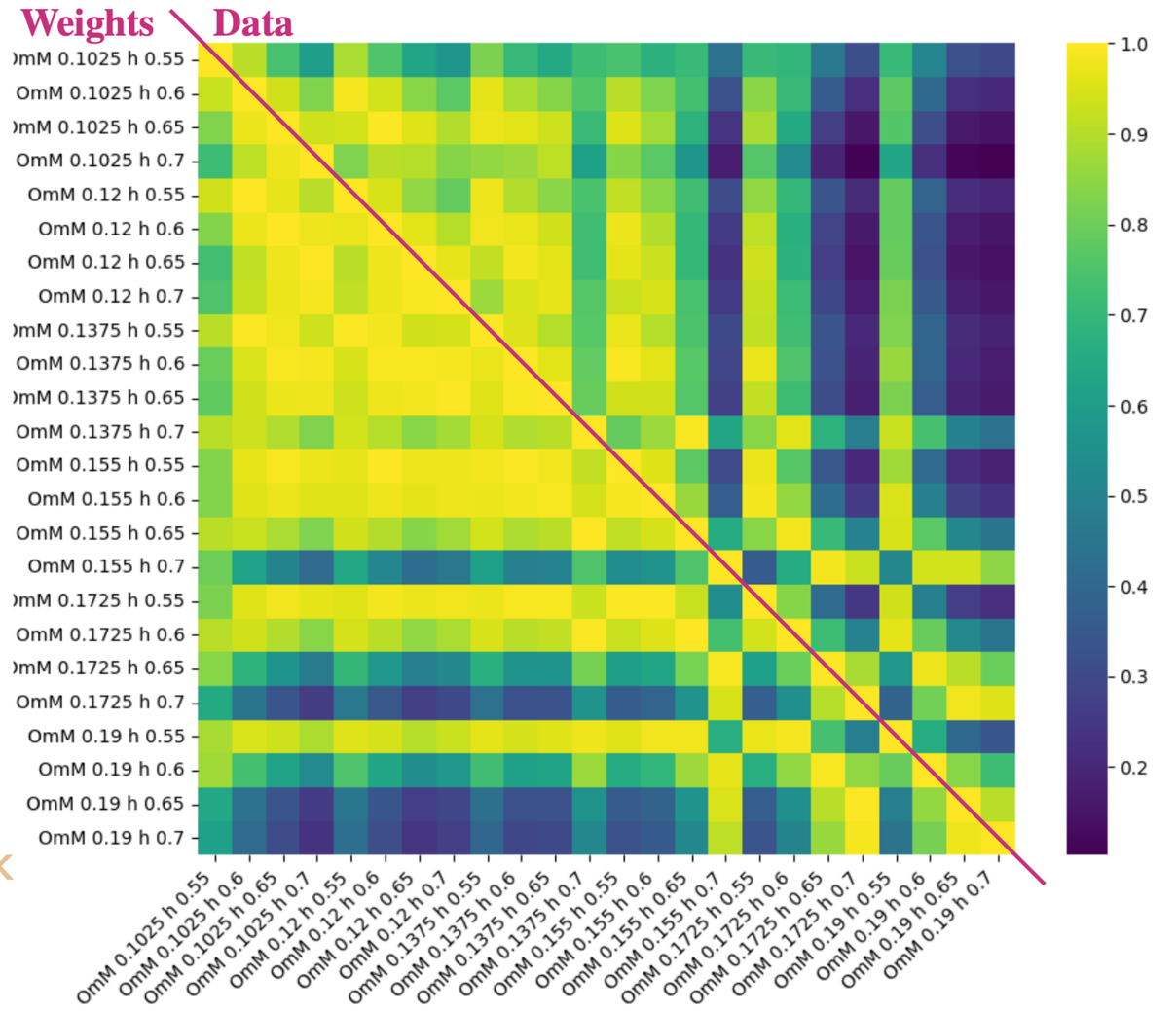


H. Gadirov, J. B. T. M. Roerdink, and S. Frey, "FLINT: Learning-based Flow Estimation and Temporal Interpolation for Scientific Ensemble Visualization,", , TVCG, 2025,

HyperFLINT



96% triplet loss accuracy



Hypernetwork

- small network generating weights for large network
- allows conditioning on simulation parameters

→ compact encoding of behavior in parameter space

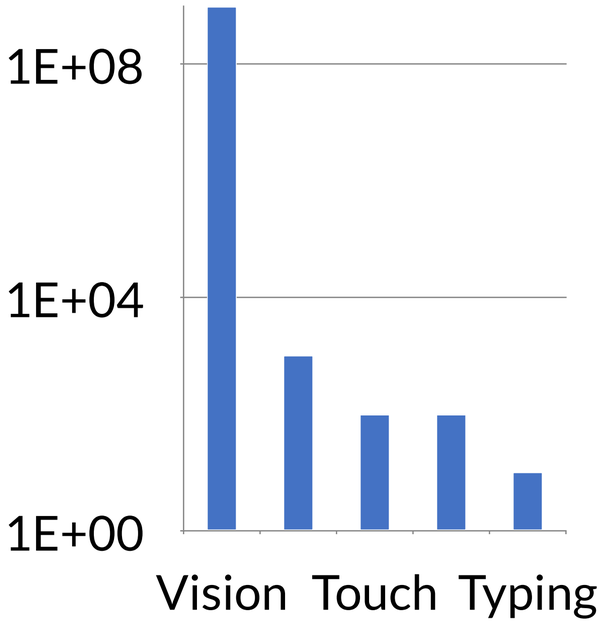
H. Gadirov, Q. Wu, D. Bauer, K.-L. Ma, J. B. T. M. Roerdink, and S. Frey, "HyperFLINT: Hypernetwork-based Flow Estimation and Temporal Interpolation for Scientific Ensemble Visualization," EuroVis 2025.

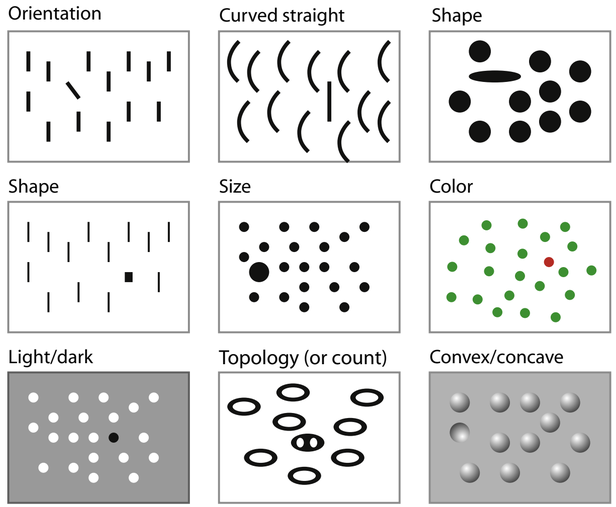
Summary

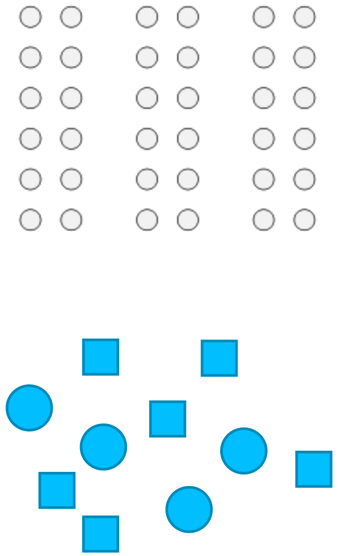
Take home messages

Summary: Large Data Visualization

- vis: most efficient data communication channel → humans

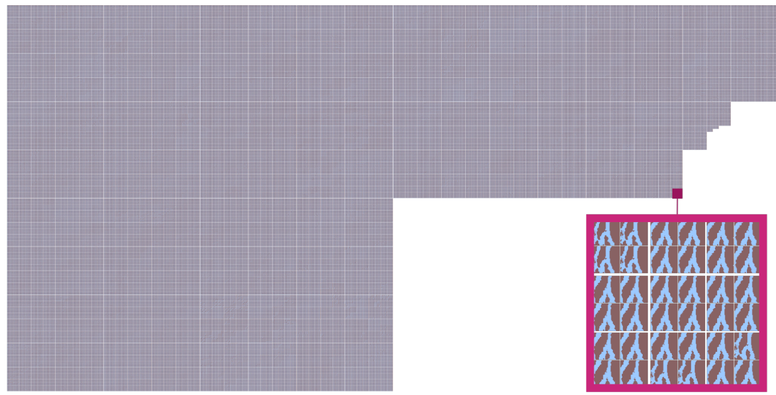






Summary: Large Data Visualization

- vis: most efficient data communication channel → humans
- “large” data: resolution, memory footprint, #data points, …



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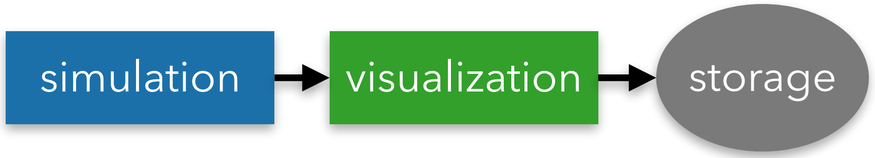

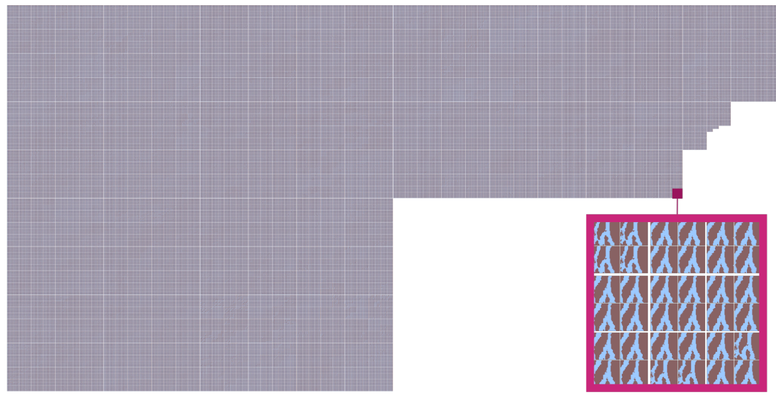

Summary: Large Data Visualization

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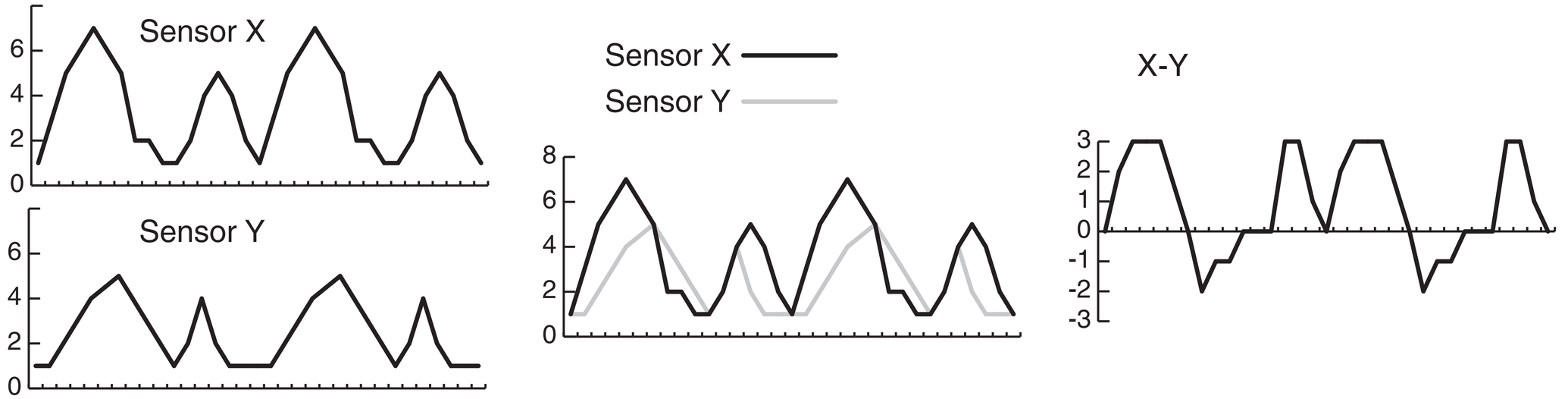
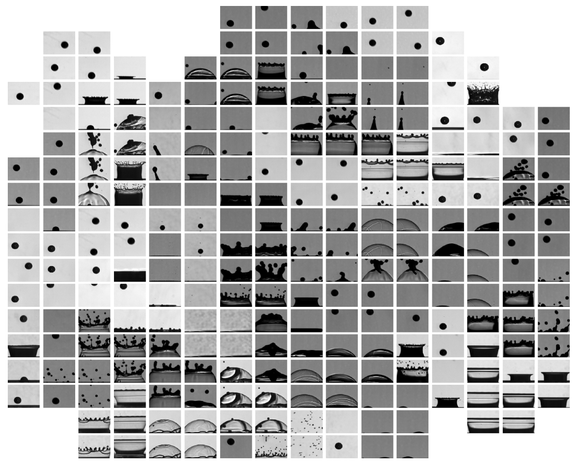
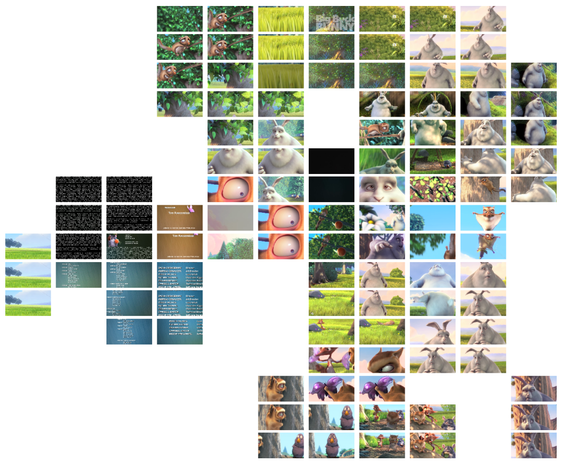
- "large" data: memory footprint, resolution, #data points, …
  - comparative visualization: identify patterns / clusters / outliers
    - designs: juxtaposition, superimposition, explicit encoding

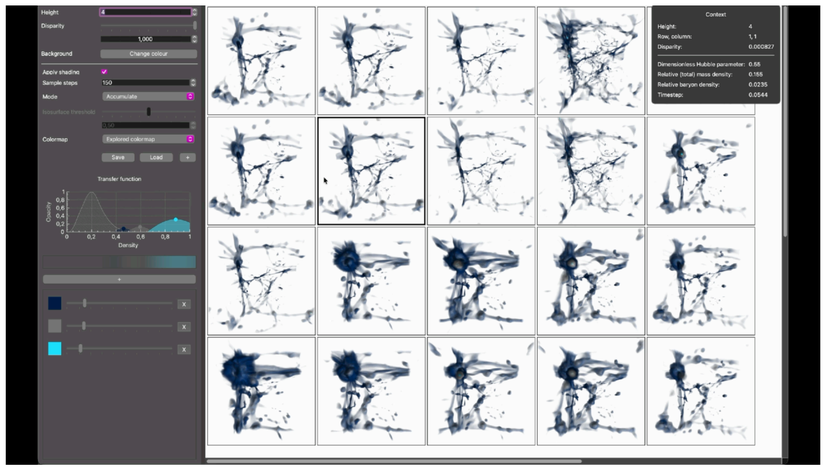


Summary: Large Data Visualization

- vis: most efficient data communication channel → humans
- “large” data: memory footprint, resolution, #data points, …
  - comparative visualization: identify patterns / clusters / outliers
    - designs: juxtaposition, superimposition, explicit encoding
      - scalable grids (on-the-fly generation, hierarchical, dynamic)

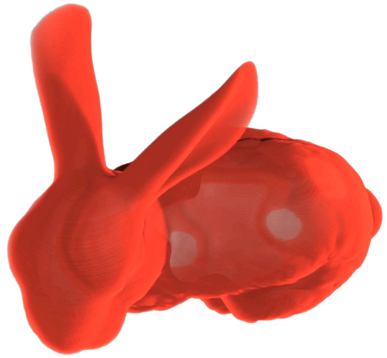


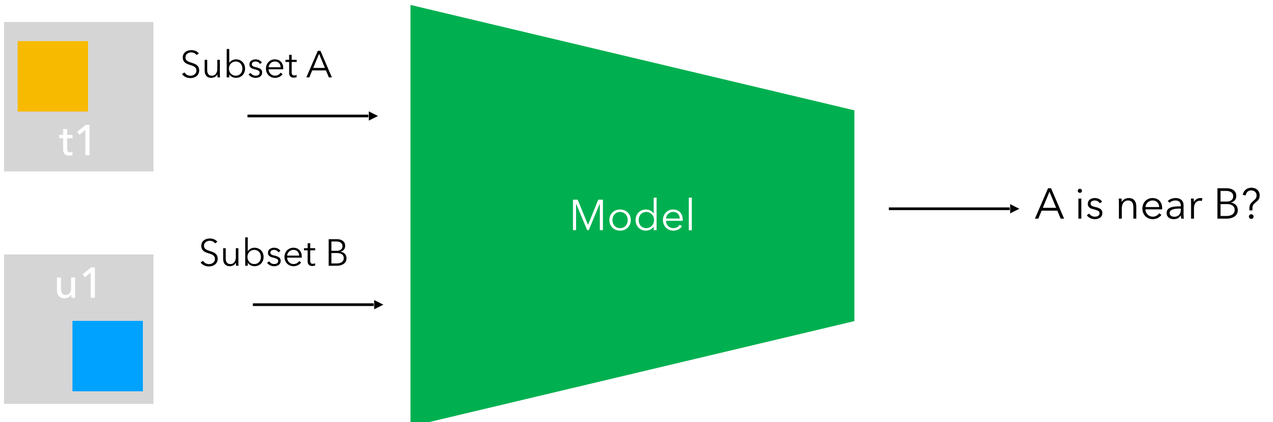




Summary: Large Data Visualization

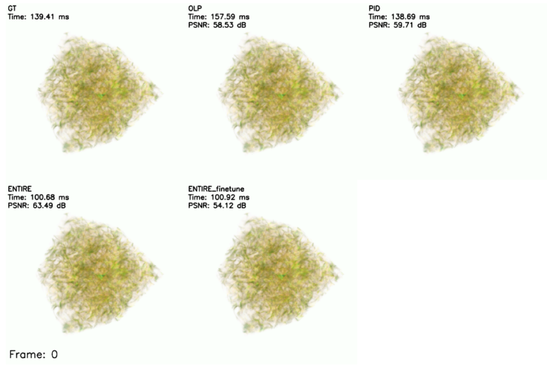
- vis: most efficient data communication channel → humans
- "large" data: memory footprint, resolution, #data points, …
  - comparative visualization: identify patterns / clusters / outliers
    - designs: juxtaposition, superimposition, explicit encoding
      - scalable grids (on-the-fly generation, hierarchical, dynamic)
    - similarity quantification crucial (element-based, flow-based, learning-based)

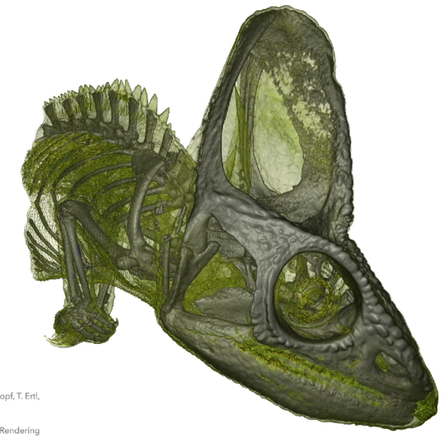




Other related large data vis. we work on …

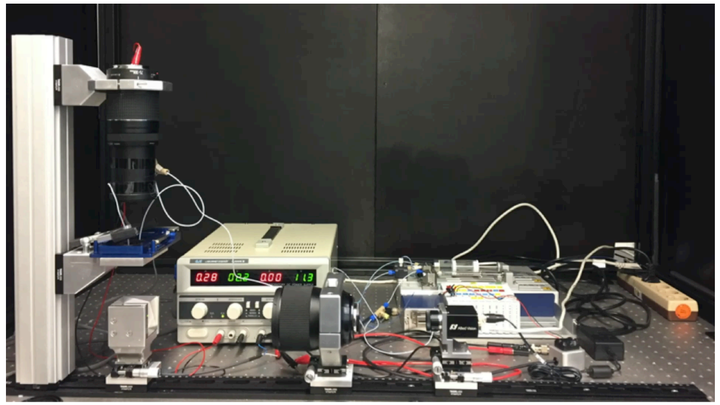
performance
(prediction, acceleration)

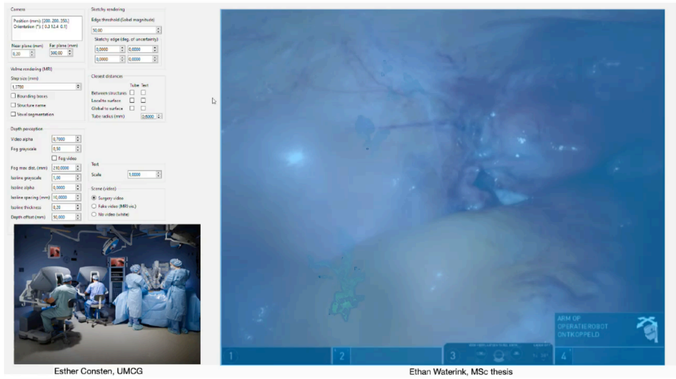




V. Bruder, C. Schulz, R. Bauer, S. Frey, D. Weiskopf, T. Ertl, Voronoi-Based Foveated Volume Rendering , EuroVis 2019 **(Best Short Paper)**, https://github.com/vbruder/FoveatedVolumeRendering

domain-specific visualization
(medical, natural sciences)

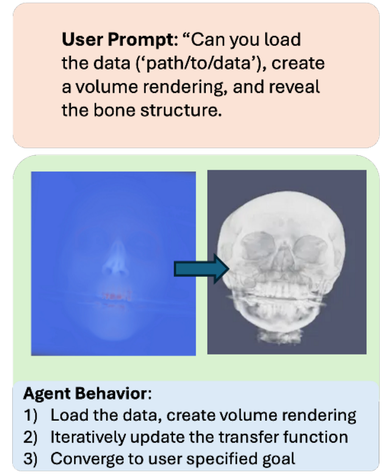


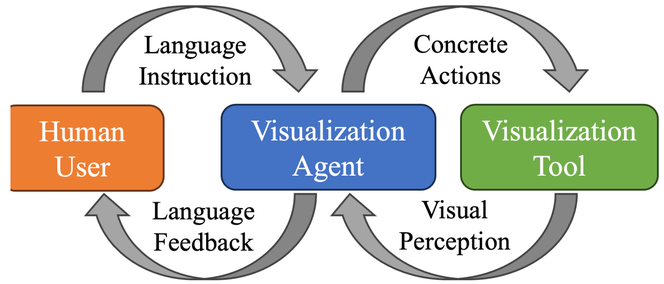


Esther Consten, UMCG

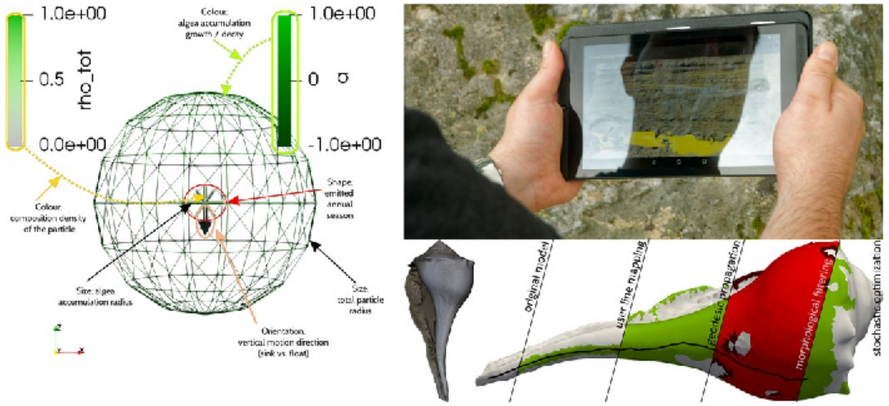
Ethan Waterink, MSc thesis

MLLM-based visualization

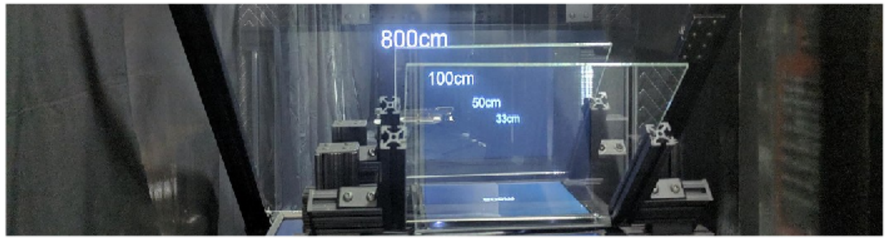




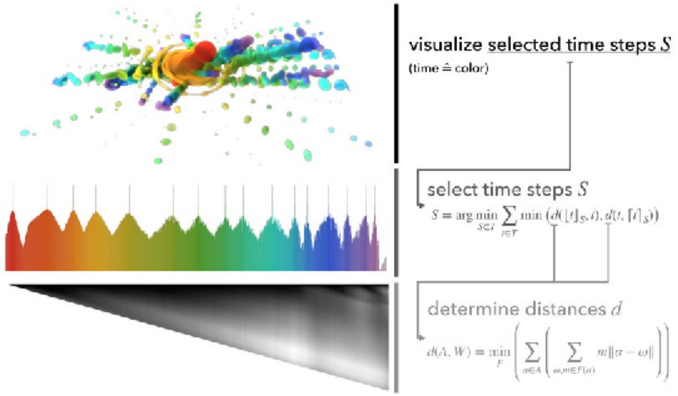
Scientific Visualization and Computer Graphics



Visualization- and Interaction Design



Perceptually optimized display systems



Optimization of scientific large data visualization

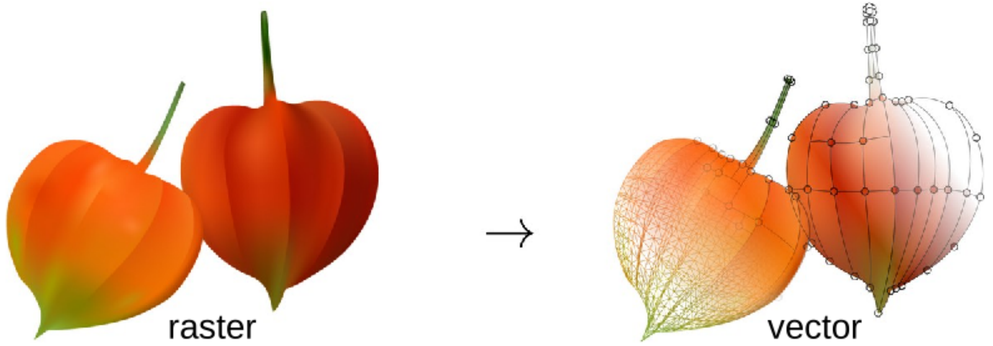


Image vectorization (fully scalable, sparse, easy to edit)

General topics and interests

- Shape and image modelling and processing
- Big data and high-performance visualization
- Optimization and machine learning
- Visual perception
- VR/AR display technology, eye tracking
- Data-driven method & system configuration
- Computer aided design and analysis
- Medical data, astronomy, …

Steffen Frey

Jiří Kosinka

Jos Roerdink

Alex Telea